

Optimal Control of the Landing Transition Phase for a Tiltrotor UAV

A thesis submitted in part fulfilment of the degree of
Bachelor of Engineering (Honours)

by
Fraser Miller
U6668747

Supervisor: Matthew Hampsey

Examiner: Robert Mahony



**Australian
National
University**

College of Engineering and Computer Science
The Australian National University

October 2025

This thesis contains no material which has been accepted for the award of any other degree or diploma in any university. To the best of the author's knowledge, it contains no material previously published or written by another person, except where due reference is made in the text.

Fraser Miller
31 October 2025

Acknowledgements

I would like to take this opportunity to show my appreciation to my Supervisor: Matt, and Examiner: Rob for their guidance and support over the year, both with this project and improving as an engineer.

Abstract

This thesis presents an initial investigation into the implementation of optimal control to improve the landing transition efficiency of the ARACE Griffin Pro - a tiltrotor hybrid UAV. The research focuses on developing a longitudinal dynamic model of the Griffin Pro that accounts for complex aerodynamics effects, including the modelling of aerodynamic coefficients in post-stall flight, estimating rotor-induced drag, and a prediction of rotor power consumption using momentum theory.

Using the developed model, a trajectory optimisation problem and linear-quadratic regulator (LQR) controller were designed to minimise energy consumption while maintaining aircraft safety during the landing phase. Simulation results demonstrated a significant reduction in energy use compared to the existing landing control system, highlighting optimal control as a promising approach for improving landing efficiency. The outcomes of this work provide a strong foundation for future development and potential physical implementation of optimal control in the Griffin Pro UAV.

Contents

Acknowledgements	i
Abstract	ii
List of Figures	vi
List of Tables	1
1 Introduction	2
1.1 Background	2
1.2 Research Context	3
1.3 Research Scope	3
2 Problem Analysis	5
2.1 Flight Data Analysis	5
2.1.1 Segment 1: Glide	6
2.1.2 Segment 2: Slow and Hover	7
2.1.3 Segment 3: Initial Descent	7
2.1.4 Segment 4: Descend and Land	7
2.1.5 Current Control Challenges	7
2.2 Problem Definition	8
2.2.1 Project Assumptions	8
2.2.2 Modelling Objectives	8
2.2.3 Aircraft Constraints	8
2.2.4 Trajectory Goals	9
2.2.5 Controller Objectives	10
3 Literature Review	11
3.1 The Unmanned Aerial Vehicle	11
3.1.1 Transition Mode	12
3.2 Aircraft Modelling	12
3.2.1 Fixed-Wing Modelling	12
3.2.2 Rotor Dynamics	14
3.2.3 Rotor Power	15
3.2.4 Aircraft Dynamics	15
3.3 Optimal Control	15
3.3.1 Trajectory Generation	16
3.3.2 Closed-Loop Control	17

4	Aircraft Modelling	19
4.1	Aeroplane Forces	20
4.1.1	Lift and Drag Models	21
4.2	Quadrotor Forces	22
4.2.1	Rotor Power	23
4.3	Aircraft Dynamics	24
5	Optimal Control	25
5.1	Trajectory Generation	25
5.1.1	System Constraints	25
5.1.2	State and Input Constraints	26
5.1.3	Collocation Method	27
5.1.4	Initial Guess	27
5.1.5	The Trajectory Optimisation Solution	28
5.1.6	Open-Loop Simulation	29
5.2	The Linear-Quadratic Regulator Feedback Controller	30
5.2.1	Error-State Representation	31
5.2.2	Feedback Gain	31
5.2.3	Successful Feedback Control	32
6	Results	34
6.1	Future Works	35
6.1.1	Model Verification	35
6.1.2	Wind Modelling	35
6.1.3	Lateral Dynamics Analysis	35
6.1.4	Ardupilot Implementation	36
6.2	Conclusion	36
	Bibliography	38
A	Appendix A: Griffin Pro Flight Data	41
B	Appendix B: Rotor Power Derivation	46
C	Appendix C: Disturbance Simulation Graphs	48

List of Figures

1.1	ARACE Griffin Pro	2
2.1	Flight F77 flight circuit	5
3.1	Flat Plate Theory approximation	13
4.1	Griffin Pro aircraft model	19
4.2	Blended lift coefficient	22
4.3	Blended drag coefficient	22
5.1	Open-loop trajectory visualisation	28
5.2	Open-loop trajectory states and inputs	29
5.3	Open-loop simulation	30
5.4	Feedback controller block diagram	30
5.5	Closed-loop state representation	33
5.6	Closed-loop input control	33
5.7	Closed-loop power and energy consumption	33
A.1	Battery measurement during flight F77	41
A.2	Desired altitude measurement during flight F77	41
A.3	Groundspeed measurement during flight F77	42
A.4	Battery measurement during flight F78	42
A.5	Desired altitude measurement during flight F78	43
A.6	Groundspeed measurement during flight F78	43
A.7	Battery measurement during flight F79	44
A.8	Desired altitude measurement during flight F79	44
A.9	Groundspeed measurement during flight F79	45
C.1	Sim 0 states	48
C.2	Sim 0 inputs	48
C.3	Sim 0 energy consumption	48
C.4	Sim 1 states	49
C.5	Sim 1 inputs	49
C.6	Sim 1 energy consumption	49
C.7	Sim 2 states	50
C.8	Sim 2 inputs	50
C.9	Sim 2 energy consumption	50
C.10	Sim 3 states	51
C.11	Sim 3 inputs	51
C.12	Sim 3 energy consumption	51
C.13	Sim 4 states	52

C.14 Sim 4 inputs	52
C.15 Sim 4 energy consumption	52
C.16 Sim 5 states	53
C.17 Sim 5 inputs	53
C.18 Sim 5 energy consumption	53
C.19 Sim 6 states	54
C.20 Sim 6 inputs	54
C.21 Sim 6 energy consumption	54
C.22 Sim 7 states	55
C.23 Sim 7 inputs	55
C.24 Sim 7 energy consumption	55
C.25 Sim 8 states	56
C.26 Sim 8 inputs	56
C.27 Sim 8 energy consumption	56

List of Tables

2.1	Interpolated Griffin Pro flight data	6
2.2	Dynamical system states and inputs	9
4.1	Griffin Pro model parameters	20
4.2	Lift and drag coefficient parameters	22
6.1	Disturbance simulations data	37

Introduction

Hybrid unmanned aerial vehicles (UAVs) were introduced to modern aerospace operations as aircraft capable of vertical take-off and landing (VTOL) while maintaining efficient long-range flight. They combine the inherent advantages of multirotor and fixed-wing UAVs while effectively eliminating the disadvantages of each UAV type in one aircraft [14, 15]. This versatility allows for a broad range of applications, with the Australian National University (ANU) currently incorporating a tiltrotor hybrid UAV known as the ARACE Griffin Pro in various surveillance projects. However, the operation of this aircraft is not without challenges; the complex, nonlinear interactions between aerodynamic lift and rotor thrust are not fully modelled for the Griffin Pro, leading to significant control difficulties particularly during transitions between forward-flight and hover/VTOL.

1.1 Background

The Griffin Pro is a subclass of hybrid UAV known as a quad-tiltrotor (Figure 1.1). The vehicle incorporates four rotors configured around the aircraft's centre of mass that are capable of tilting 90° between a vertical configuration, enabling VTOL, and a horizontal configuration in which the aircraft uses the front rotors for thrust during forward-flight.

The limited understanding of the aerodynamic coupling between the rotors and the fixed-wing components during the transition phase for this relatively new aircraft [28], has led to the use of a simplistic and energy intensive landing transition control. The current control approach (developed by Ardupilot) relies on induced drag generated by the rotors to gradually slow the aircraft's momentum to a stable hover before initiating VTOL descent. This gradual transition allows the control system to stabilise any potential aerodynamic instabilities as aerodynamic loads shift from the fixed-wing control surfaces to the rotors. However, this process incurs a substantial energy cost thereby reducing the potential mission duration.

Research on tiltrotor modelling and control has demonstrated success [32], with studies showing



Figure 1.1: Image of the ARACE Griffin Pro. Image credit:[3].

that automating the transition phase by developing systems that control the dynamics during this highly nonlinear period can reduce flight energy consumption [17]. This highlights the opportunity to develop a tailored landing transition control system for the Griffin Pro to improve landing energy efficiency, ultimately benefiting the ANU and other Griffin Pro stakeholders by potentially increasing mission duration of the aircraft.

1.2 Research Context

By focusing on the longitudinal dynamics of the Griffin Pro, this project investigates the need for improved aircraft modelling and a control framework that enables a stable and energy efficient transition between forward-flight and hovering. Optimal control is identified as a promising approach to improve this system, leading to the following research question:

- How can optimal control be implemented to improve the landing transition energy efficiency of the Griffin Pro UAV?

This initial investigation aimed to develop and evaluate a control system built in MATLAB and tested in Simulink. To achieve this aim, the project followed these specific objectives:

- Analyse the current landing system to identify opportunities for improvement;
- Simplify the broader system for an initial investigation;
- Develop a longitudinal dynamic model of the ARACE Griffin Pro UAV;
- Develop a model for estimating rotor power consumption;
- Design an optimal trajectory that reduces the landing system's energy requirements while ensuring aircraft safety;
- Design a linear-quadratic regulator (LQR) controller that ensures system safety and convergence to the desired trajectory; and
- Simulate and evaluate the system's performance compared to the existing control approach.

1.3 Research Scope

To achieve these objectives, this research followed a structured methodological approach. The study began with an analysis of the existing landing transition phase to develop an understanding of the areas requiring improvement. This analysis allowed the problem to be contextualised by simplifying the broader system through the implementation of project assumptions and the development of chapter-specific goals, improving workflow and aligning tasks with the overall project goals.

A review of the existing literature was then conducted to develop the necessary understanding and tools for mathematical modelling of the Griffin Pro and optimal control implementation. The review also reinforced the unique challenges faced by hybrid UAVs and the project's underlying assumptions.

Following this review, the aircraft's model was produced, addressing the challenges of dynamically modelling the aerodynamic, propulsive, and gravitational forces acting on the aircraft in motion. These challenges included modelling of the aerodynamic coefficients beyond available aerofoil data,

computing the resulting aerodynamic forces, rotor thrust, and induced drag, and estimating the power required to generate these forces.

Once the aircraft model was established, optimal control was implemented. This process was split into two stages: firstly designing the trajectory optimisation problem (incorporating aircraft limitations and the trajectory goals defined during the problem analysis) and secondly designing the LQR to ensure aircraft safety during simulation and the convergence of the system to the desired trajectory.

Various disturbed trajectories were then simulated to identify controller limitations and estimate the effect of the control system on energy reduction. Finally, potential avenues for project extension and integration into a physical system were discussed.

Problem Analysis

To understand the existing system and formulate the project problem, the project was provided with flight logs containing data from three short flights (flights F77, F78, and F79) conducted by the ARACE Griffin Pro UAV. Each flight followed the same flight path, guided by thirteen waypoints (Waypoints 1 - 13), around a short circuit multiple times before returning to the landing position, as shown in Figure 2.1. When the aircraft was instructed to return to the initial position, it glided through Waypoints 9 - 12 to descend and decelerate before transitioning into a hover state at Waypoint 13, where the aircraft's VTOL system allowed it to descend and land.

As expected, during gliding flight the aircraft is highly energy efficient; therefore, the landing transition energy analysis began when the aircraft reached Waypoint 12 and levelled out into forward flight. In forward flight, the lateral dynamics are stabilised near zero, while the longitudinal dynamics are dominate the system behaviour. Using MAVExplorer, flight data from the flights were visualised as graphs that were interpolated to produce Table 2.1. The flight data graphs are presented in Appendix A.

According to the Griffin Pro specification sheet, the UAV uses a 22.2V, 30Ah Lithium-ion battery with a total energy storage capacity of 666Wh [6]. From the fight data graphs it was determined that during cruise flight the aircraft consumes approximately 286W, while in idle, the onboard systems consume approximately 12W. The idle power was subtracted from the energy values presented in Table 2.1.

2.1 Flight Data Analysis

Beginning at Waypoint 12, the flight path was divided into four segments defined by five points along the trajectory, allowing the various flight states during the landing transition to be analysed in terms of their energy consumption. The points being at Point 0 and are defined as follows:

0. Waypoint 12,
1. the aircraft is 50m laterally from the target,

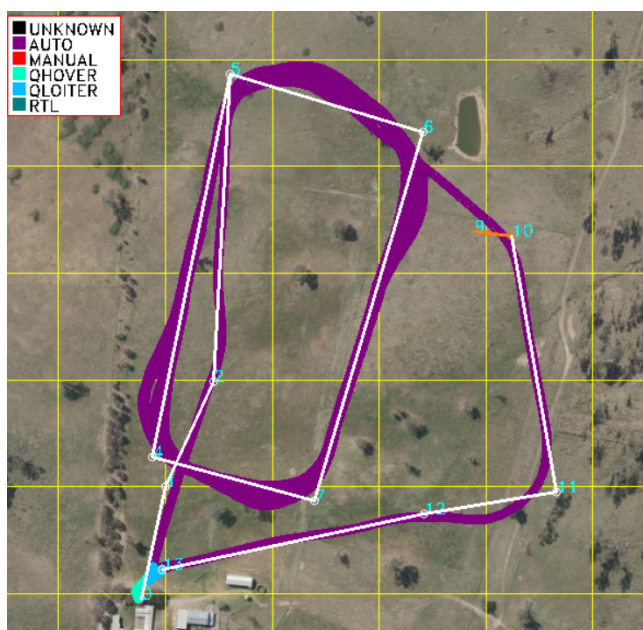


Figure 2.1: Flight F77 depicting the flight circuit flown by the Griffin Pro.

(0) Waypoint 12	F77	F78	Average(F77,F78)	F79	3 Flight Average
Time (s)	0	0	0	0	0
Horizontal displacement (m)	252	241	246.5	243	245.333
Vertical displacement (m)	45.6	47.1	46.35	59.3	50.667
Horizontal velocity (m/s)	21.1	20.3	20.7	23.0	21.467
Total Energy (Wh)	0	0	0	0	0
(1) 50m Horizontal from Target	F77	F78	Average(F77,F78)	F79	3 Flight Average
Time (s)	13.5	11	12.25	10.9	11.8
Vertical displacement (m)	46.6	47.8	47.2	58.6	51
Horizontal velocity (m/s)	15.5	17.2	16.35	13.7	15.467
Total Energy (Wh)	1.04	1.26	1.15	0.45	0.917
Segment Energy (Wh)	1.04	1.26	1.15	0.45	0.917
(2) Begin Descent	F77	F78	Average(F77,F78)	F79	3 Flight Average
Time (s)	24	23	23.5	16	21
Vertical displacement (m)	50.2	48	49.1	58.3	52.167
Total Energy (Wh)	4.64	4.65	4.645	1.27	3.52
Segment Energy (Wh)	3.6	3.39	3.495	0.82	2.603
(3) 20m Below Initial Altitude	F77	F78	Average(F77,F78)	F79	3 Flight Average
Time (s)	33	33	33	25	30.333
Vertical displacement (m)	25.6	27.1	26.35	39.3	20.667
Total Energy (Wh)	7.82	8.37	8.095	4.53	6.907
Segment Energy (Wh)	3.18	3.72	3.45	3.26	3.387
(4) Surface Touchdown	F77	F78	Average(F77,F78)	F79	3 Flight Average
Time (s)	77	75	76	98	83.333
Vertical displacement (m)	0	0	0	0	0
Total Energy (Wh)	21.14	24.31	22.725	32.94	26.13
Segment Energy (Wh)	14.32	15.94	15.13	28.4	19.553

Table 2.1: Landing transition data interpolated from the Griffin Pro F77, F78, and F79 flight logs.

2. the aircraft is in hover just before beginning its descent,
3. the aircraft is 20m below the altitude recorded at Point 0, and
4. the aircraft has just landed on the surface.

Among the three flights, flights F77 and F78 were determined to be very similar, exhibiting similar durations and states at each points. In contrast, due to a stronger tailwind, flight F79 began at Waypoint 12 travelling approximately $2m/s$ faster, with an altitude about $14m$ higher than those recorded in F77 and F78.

2.1.1 Segment 1: Glide

Between Waypoint 12 and Point 1, The Griffin Pro travels approximately $200m$. During this segment, the aircraft primarily glides while decelerating and completing its descent in preparation for

landing. Notably, the energy expenditure during this segment appears proportional to the reduction in velocity. This relationship is evident in flight F79, in which the aircraft decelerated by approximately $9m/s$ while consuming only 35 – 45% of the energy used in Segment 1 of flights F77 and F78, which slowed by approximately $5.5m/s$ and $3m/s$, respectively.

2.1.2 Segment 2: Slow and Hover

Between Points 1 and 2, the aircraft reduces forward momentum and transitions into a stable hover in preparation for the VTOL descent. During this phase, the control system reduces the aircraft momentum by generating induced drag through the rotors. Consequently, when the aircraft velocity is higher at Point 1 (as observed in flights F77 and F78) the system expends more energy to arrest the aircraft's momentum. In contrast, flight F79, which began this segment with less velocity, required less energy to stabilise into hover. This segment also includes an inefficient stabilisation period at Waypoint 13, during which the aircraft maintains minimal forward velocity by exhibits a short delay before initiating descent.

2.1.3 Segment 3: Initial Descent

Point 3 was defined as $20m$ below the aircraft's altitude at Waypoint 12, a height chosen to ensure a stable and consistent descent without the disturbance of any landing preparation control measures. The energy consumption during this segment was found to be similar across all three flights when normalised by segment duration. This consistency indicates a potential opportunity for improved efficiency by reducing or bypassing this routine descent region.

2.1.4 Segment 4: Descend and Land

Analysis of Segment 4 confirmed that the VTOL descent represents the largest energy demand of the landing phase. In flights F77 and F78, the average power and energy consumption were approximately $1.27kW$ and $0.57Wh/m$, respectively. In contrast, flight F79 (featuring a descent distance approximately 50% greater and a descent time approximately 70% longer) recorded an average power and energy consumption of $1.4kW$ and $0.72Wh/m$, respectively. While stabilisation near touchdown remains a necessary safety feature of the system, Segment 4 highlights that minimising the descent distance and duration above this stabilisation region would significantly improve energy efficiency.

2.1.5 Current Control Challenges

The segmented landing strategy identified in the Griffin Pro arises from the modelling complexity inherent in nonlinear tiltrotor dynamics. This complexity stems primarily from aerodynamic coupling between the fixed-wing control surfaces and the rotors, which can induce instability during transition, as well as from the challenges associated with accurately modelling rotor lift and induced drag. Consequently, without an appropriate method for capturing these dynamics, the system transitions gradually into a hover, allowing continuous stabilisation of the coupling effects before VTOL descent. To address these challenges and progress towards the project objectives, it was essential to develop a high-fidelity dynamic model of the Griffin Pro's capable of representing the aerodynamic interactions and nonlinear behaviour that currently causes operation challenges.

2.2 Problem Definition

Analysis of the flight data revealed that improving the energy efficiency of the landing transition requires trajectory design to satisfy four key objectives:

- Minimise the distance and duration of the landing transition;
- Minimise the distance and duration of the VTOL descent;
- Reduce rotor energy consumption; and
- Ensure that the vehicle maintains flight stability and safety throughout the transition phase.

Achieving these objectives requires three core capabilities within the project: a dynamic model that accurately represents aircraft behaviour throughout the landing transition, a method to estimate rotor power and energy consumption, and an understanding of the vehicle's operational limits to maintain safe flight conditions during transition.

2.2.1 Project Assumptions

To constrain the scope of this initial investigation, several simplifying assumptions were made. The project does not consider the effects of wind, as examined in [27], however, wind modelling would be essential for physical implementation. The primary simplification concerns the restriction of the dynamics model to the longitudinal frame. This approach, consistent with previous studies [11, 27, 33], was justified by the dominance of longitudinal dynamics during landing. In this phase, the lateral control objective is limited to maintaining stability rather than influencing the descent trajectory. It is assumed that the current control system of the Griffin Pro adequately stabilise lateral dynamics, while the control system developed in this work overrides only the longitudinal control during the landing transition, before returning control back to the existing controller for the remaining VTOL descent and landing.

2.2.2 Modelling Objectives

Under these assumptions, the project aimed to develop a model capable of meeting the trajectory design requirements by:

- Predicting complex aerodynamic phenomena, including rotor induced drag;
- Estimating rotor power and energy consumption; and
- Representing the aircraft as a nonlinear dynamic system:

$$\dot{\mathbf{Y}}(t) = f(\mathbf{Y}(t), \mathbf{u}(t), t),$$

where $f : \mathbb{R}^{10} \times \mathbb{R}^4 \rightarrow \mathbb{R}^{10}$; $\mathbf{Y}(t) \in \mathbb{R}^{10}$ represents the ten states of the aircraft and $\mathbf{u}(t) \in \mathbb{R}^4$ is the input of the aircraft, as defined in Table 2.2.

2.2.3 Aircraft Constraints

To develop a feasible trajectory, the aircraft constraints were defined for three main reasons: to replicate the physical limitations of the aircraft, to ensure operation safety, and to prevent actuator saturation. Some physical limits were easily identified, such as the tilt range of the front rotors and

State	Description	State	Description
x	Horizontal displacement (m)	$\dot{x}$	Horizontal velocity (m/s)
z	Vertical displacement (m)	$\dot{z}$	Vertical velocity (m/s)
θ	Aircraft pitch angle (rad)	$\dot{\theta}$	Aircraft pitch rate (rad/s)
$\bar{\omega}_{fr}$	Speed of the front rotors (rpm)	$\bar{\omega}_{rr}$	Speed of the rear rotors (rpm)
ξ	Tilt angle of the front rotors (rad)	ε	Tilt angle of the elevator (rad)
Input	Description	Input	Description
$\dot{\bar{\omega}}_{fr}$	Acceleration of the front rotors (rpm/s)	$\dot{\bar{\omega}}_{rr}$	Acceleration of the rear rotors (rpm/s)
$\dot{\xi}$	Tilt rate of the front rotors (rad/s)	$\dot{\varepsilon}$	Tilt rate of the elevator (rad/s)

Table 2.2: Dynamical system states $\mathbf{Y}(t)$, and inputs $\mathbf{u}(t)$ for the Griffin Pro model. Where $\mathbf{Y}(t) = [x \ z \ \theta \ \dot{x} \ \dot{z} \ \dot{\theta} \ \bar{\omega}_{fr} \ \bar{\omega}_{rr} \ \xi \ \varepsilon]^T$, $\mathbf{u}(t) = [\dot{\bar{\omega}}_{fr} \ \dot{\bar{\omega}}_{rr} \ \dot{\xi} \ \dot{\varepsilon}]^T$.

elevator, given by $\xi \in [0, \frac{\pi}{2}]$ and $\varepsilon \in [-\frac{\pi}{4}, \frac{\pi}{4}]$, as well as the rotor speed cap of 10000 rpm. However, most limitations were defined to ensure the aircraft's safety.

Given that constraints cannot be implemented in an LQR controller, the limit values were deliberately set conservatively. This ensured that if the controller near or slightly beyond these bounds, the system's structural integrity and flight safety would not be compromised. To further safeguard the aircraft the rotor speed was restricted to positive values to prevent the production of downward thrust, by $\bar{\omega}_{fr} = \bar{\omega}_{rr} \in [0, 10000]$. Additionally, acrobatic manoeuvres were prevented by limiting the pitch angle and pitch rate: $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$ and $\dot{\theta} \in [-\frac{\pi}{4}, \frac{\pi}{4}]$. Restricting these states to $\pm 45^\circ$ provides a logical bound to prevent the LQR controller from commanding aggressive or acrobatic manoeuvres.

Finally, bounds were imposed on the rates of change of the control states to avoid actuator saturation and prevent "bang-bang" control. These were applied by limiting the inputs as the derivatives of the control states by $\dot{\bar{\omega}}_{fr} = \dot{\bar{\omega}}_{rr} \in [-5000, 5000]$, $\dot{\xi} \in [-\frac{\pi}{4}, \frac{\pi}{4}]$, and $\dot{\varepsilon} \in [-\frac{\pi}{4}, \frac{\pi}{4}]$.

2.2.4 Trajectory Goals

With a representative model of the Griffin Pro, defined safety constraints, and clear design objectives, specific goals were developed to achieve the desired performance improvements. The initial state of the trajectory must represent the aircraft conditions at Waypoint 12 when the Griffin Pro is in cruise flight, while the final state corresponds to a stable hover, allowing handover to the existing control system.

To reduce the distance and duration of the transition to hover and the VTOL descent, the trajectory was designed to blend the two phases. Rather than performing a 250m transition followed by a $> 40m$ VTOL descent, the blended trajectory begins 50m horizontally from the target and reduces the VTOL descent by 20m, aiming to complete both in a single continuous manoeuvre.

In the existing system, the aircraft requires approximately 30s to travel from Waypoint 12 to the descent point 20m below its initial altitude. This target for this trajectory was to achieve the transition between these points in 7s - a significant and realistic reduction with the implementation of optimal control. These goals are expected to substantially reduce energy consumption during landing; however, to realise this in practice, a feedback controller was required.

2.2.5 Controller Objectives

The controller was implemented primarily to ensure that the aircraft converges to the desired trajectory while maintaining safety under all conditions. Essentially regardless of any disturbances or instabilities, upon completing the trajectory, the system could safely hand over to the existing system to safely land the vehicle. Hence, the controller was designed to place a great emphasis on finish in a hover state regardless of the aircraft position. Secondly, the controller should minimise unnecessary rotor activity to reduce extra consumption and enhance the efficiency during transition.

With these clearly defined goals for modelling, trajectory design, and control, the project methodology could be systematically followed as outlined in the following chapters.

Literature Review

Modelling hybrid UAVs and implementing optimal control systems to improve aircraft operating systems is no new problem for the aerospace community, hence the purpose of this literature review is to establish a comprehensive understanding of the research relevant to the development of an optimal longitudinal landing controller for the ARACE Griffin Pro UAV aligned with the project and chapter specific goals. Specifically, it aims to contextualise the landing problem for tiltrotor UAVs by identifying the unique challenges specific to this class of vehicle during the landing transition, to explore longitudinal modelling techniques for the successful depiction of aerodynamic and propulsive forces used in system design including estimation tools for modelling rotor power, and to investigate optimal control literature surrounding successful trajectory optimisation problem design and effective LQR controllers.

3.1 The Unmanned Aerial Vehicle

The turn of the 21st century has seen the growth of the Unmanned Aerial Vehicle (UAV) into the major aerospace market seen today. UAVs can be classified as multirotor, fixed-wing, single rotor, or hybrid UAVs [4], with each class offering a unique set of advantages and disadvantages. Rotor dynamics utilised by multirotors (quadrotor) and single rotors (helicopters) allow for a highly manoeuvrable aircraft at the cost of a limited range of use due to large energy consumptions [16]. Fixed-wing drones consume less energy than multirotors, with wing aerodynamics allowing high speed, efficient, long-range flight at the cost of the need for runways for take-off and landing [16]. Hybrid aircraft were introduced to the market to combine the long endurance and high speed of fixed-wing aircraft and the manoeuvrability and robustness arising from vertical take-off and landing (VTOL) capabilities of a multirotor, allowing a broader application to effectively eliminate individual disadvantages of each type of UAV [14, 15].

Hybrid UAVs are a diverse class consisting of tail-sitter and convertiplane UAVs. Tail-sitter aircraft take-off and land vertically on the tail of the aircraft, with the entire airframe tilting to achieve cruise flight. These aircraft are separated based on thrust transition mechanism and date back to the 1950s [28]. Conventiplane UAVs maintain the orientation of the airframe throughout flight with certain mechanisms used to transfer between VTOL and forward flight. They are separated into subclasses depending on the actuation mechanisms employed to transfer between hover and forward flight, with these subclasses being: tiltrotor, tilt-wing, rotor-wing and dual systems[28].

The aircraft supplied for this project (ARACE Griffin Pro) is a quad tiltrotor UAV. Tiltrotors actuate a pair of rotors to tilt between a vertical configuration expected in a traditional quadrotor and a horizontal state in which the rotors can function as forward thrust propellers for the aeroplane component of the aircraft during forward flight. Quad tiltrotors were described as rare in 2018 by Saeed et al. [28] compared to other tiltrotor configurations: underpinning the need for modelling and

control system development to improve the performance of these aircraft. Although these aircraft are highly capable due to the combination of rotor and fixed-wing aerodynamics and control, this combination also introduces inherent complexity due to the coupling of these systems; particularly in the transition phase between cruise flight and hover mode.

3.1.1 Transition Mode

Transition mode occurs when the aircraft is transitioning between forward flight and hover mode. During forward flight, the aircraft's fixed-wing component is the primary control surface with little influence from rotors, other than replicating propellers for forward thrust. However, during hover mode/VTOL the situation is reversed, i.e. very little aerodynamic influence arises in the wing and tail, while the vertically configured rotors are the primary producers of force. During the transition phase, when actuators tilt the rotors between configurations, the forces developed on the individual control surfaces can couple, leading to potential inefficiency and poor performance. Work to automate the transition mode has shown that developing systems that control the dynamics during this highly nonlinear period can reduce flight energy consumption [17]. This underpins the opportunity to develop a specifically tailored system that improves aircraft control during transition periods, reducing energy requirements, and improving the application of the system for the betterment of stakeholders by potential increasing mission duration of the aircraft.

3.2 Aircraft Modelling

The modelling approach in this paper builds on the approach of Roberts et al. (2009) [27] by adding tiltrotor dynamics to the simple aeroplane system to represent a Griffin Pro. A goal of modelling is to represent the dynamical system by vehicle states ($\mathbf{Y}(t) \in \mathbb{R}^m$) and inputs ($\mathbf{u}(t) \in \mathbb{R}^n$) subject to aircraft's longitudinal dynamics ($f : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \mathbb{R}^m$) with respect to time (t), given by:

$$\dot{\mathbf{Y}}(t) = f[\mathbf{Y}(t), \mathbf{u}(t), t]. \quad (3.1)$$

Roberts et al. has shown that the longitudinal dynamics of the Griffin Pro can be represented by the ten states ($m = 10$) and four inputs ($n = 4$) presented in Table 2.2[27] [32]. To model the longitudinal dynamics of the hybrid aircraft, assumptions can be drawn from Roberts' paper that model a simple aircraft as a rigid body subject to aerodynamic, propulsive, and gravitational forces within a windless environment [27].

Within this literature analysis, the aircraft modelling is separated into fixed-wing and quadrotor modelling components to develop an understanding of the existing research that allowed this project to successfully achieve its modelling goals to predict complex aerodynamics including rotor induced drag, estimate power and energy consumption of the rotor system, and represent the aircraft as a dynamical system as expressed above.

3.2.1 Fixed-Wing Modelling

The modelling of fixed-wing aircraft as done by Roberts et al. drew influence from the Wickenhieser et al. (2005) paper [31] which presents the longitudinal dynamics of a specific aircraft as the sum of aerodynamic forces and torques developed on the aircraft's control surfaces. This paper was able to conclude from its restricted longitudinal analysis that perching manoeuvres would now add a new capability to aircraft without sacrificing efficiency, developing a line of research on optimal landing

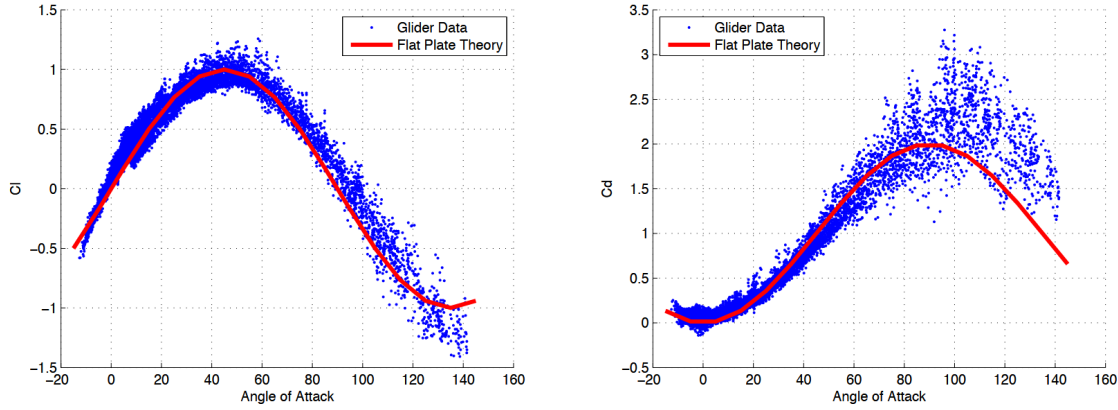


Figure 3.1: Graphs drawn for Cory and Tedrake’s paper [11] depicting Flat Plate Theory approximation with real data.

trajectories to increase energy efficiency. However, it also exposed the challenge of modelling aerofoil lift and drag coefficients during high angle-of-attack (post-stall) flight when available aerofoil data are exceeded. As post-stall flight was expected within this project’s landing transition trajectory, the development in Cory and Tedrake’s 2008 paper [11] that confirmed Flat Plate Theory is applicable for post-stall estimations of aerofoil lift and drag coefficients was a very important establishment. Flat Plate Theory is expressed by:

$$C_{L_{fp}} = 2 \sin(\alpha) \cos(\alpha), \quad C_{D_{fp}} = 2 \sin^2(\alpha), \quad (3.2)$$

where α is the angle of attack of the aircraft. The approximation is shown in Figure 3.1 which was drawing from [11]. It must be noted that Flat Plate Theory alternatives exist, as in Stengel’s (2005) *Flight Dynamics*[29], however, later work has shown Cory and Tedrake’s estimate to have more relevance in this paper [33]. The theory was further applied in the 2009 *Controllability* paper [27] as an aerofoil estimation for a general plane when creating the lift and drag models required for the control experiments. Using Flat Plate Theory for aerodynamic coefficient estimation, the paper presents equations for lift and drag forces developed on the fixed-wing control surfaces (wing and tail), that allowed this project to model the aeroplane aerodynamics. The lift and drag forces can generally be expressed by:

$$L = \frac{1}{2} \rho |V|^2 S C_L(\alpha), \quad D = \frac{1}{2} \rho |V|^2 S C_D(\alpha), \quad (3.3)$$

Where ρ is the density of air, V is the velocity of the aircraft and S is the surface area of the control surface. However, it must be noted that the paper presents a simple aircraft model, which leads to limitations in modelling implementation. These limitations arise in the sole reliance on Flat Plate Theory as a means of generating lift and drag coefficients and the papers restriction to strictly analysing the longitudinal dynamics and ignoring the lateral dynamics. Realistically, aircraft are designed with specific aerofoils; often these are non symmetrical and generate lift and drag at zero angle of attack which Flat Plate Theory fails to replicate. As such, introducing specific coefficient data, when possible, is integral to true aircraft modelling, as was required in this project. Further to this, while these papers have shown that longitudinal dynamics are primary in landing scenarios, lateral stability concerns can arise in the three degrees of freedom ignored in pure longitudinal analysis, much to the detriment of the fidelity of a model’s application. Should the inclusion of the lateral dynamics model be required for extension of this project to practical testing, Beard and McLain present a method

in their 2012 book [8]. This book provides support and avenues for model building in Matlab that was utilised for this project, including all six degrees of freedom required to model the longitudinal and lateral dynamics of a fixed-wing aircraft. The book draws upon material developed in Stengel's (2005) *Flight Dynamics* and applies it to a small UAV, the Aerosonde, to produce a stable model capable of adhering to specified flight requirements via PID control. Within the book is the introduction of a sigmoid function for modelling aerodynamic coefficients that represent specified aerofoil data while allowing for post-stall modelling with Flat Plate Theory. This concept improved the fidelity of this project's model and other models that were expected to experience high angle-of-attack flight, as seen in [33].

3.2.2 Rotor Dynamics

To produce a model capable of predicting rotor dynamics, the project explored literature on rotor dynamics. Rotors dynamics are inherently complex to model due to the aerodynamic phenomena of induced drag and blade flapping [25]. Initially, these nonlinear occurrences were ignored in rotor modelling, but advances in computational abilities have shown that the inclusion of these phenomena improves the overall robustness of rotor models [23]. Accurate methods for modelling these complex dynamics have been developed in blade element theory (BET)[13], momentum theory (MT)[7], and the combination of the two: blade element momentum theory (BEM) [7]. Blade element theory attempts to estimate the thrust of a propellor or rotor by dividing each blade into segments expressed as blade elements. The theory treats each individual element as a two dimensional aerofoil which allows for local aerodynamic forces to be developed at each segment and the overall rotor forces expressed as their summation. This allows for an accurate depiction of the rotors, as they often change in size, shape, and thickness throughout the length of the blade [13]. Bangura et al. (2016) explains modelling momentum theory and BEM in their paper [7], drawing material from Leishman's (2006) book [21]. Momentum theory considers a rotor as an actuator disc with accelerated air forming a stream tube. Blade element momentum theory expands upon BET by incorporating momentum theory equations for the conservation of axial and tangential momentum within a streamtube to alleviate some of the difficulties arising in calculating induced velocities at the rotor and is the most popular tool for predicting aerodynamic forces on a rotating blade [7, 24].

Although using one of these methods above would be desirable for a strong representation of rotor aerodynamics, there are inherent complexities with each method. Modelling BET and BEM require complex rotor blade parameters and can be very challenging to implement in a system, and while momentum theory is far simpler, it relies upon the avoidance of axial decent flight regimes that may be replicated in standard flight, leading to an unstable system. In addition to this, the modelling equations for momentum theory are under-expressed as there are more unknown variables than equations to express them. However, it is supported in Bangura's et al. paper that the aerodynamics of rotors can be estimated from BEM [7, pp. 21-24] by:

$$L_r = \frac{1}{4} N_b \rho c \bar{\omega}^2 C_T R_r^3, \quad D_r = L_r \mu, \quad (3.4)$$

where μ is the advanced ratio of the rotor specified by:

$$\mu = \frac{V_h}{R_r \bar{\omega}}, \quad (3.5)$$

C_T is the thrust coefficient, $\bar{\omega}$ is the rotor speed (rpm), V_h is the airflow velocity parallel to the rotor blades, N_b is the number of rotor blades, c is the rotor chord which can be assumed constant, and R_r is the radius of the rotor disc.

3.2.3 Rotor Power

With estimations for rotor lift and drag, the momentum theory system given by:

$$L_r = 2\rho\pi R_r^2 v_z^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (3.6)$$

$$D_r = 2\rho\pi R_r^2 v_x^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (3.7)$$

$$P_r = L_r(v_z^i - V_{vr}) + D_r(v_x^i + V_{hr}), \quad (3.8)$$

was solvable to estimate the rotor power. This was a very important identification for the project, as it allowed for system energy measurements and comparison between the existing control and the control model produced in this project.

3.2.4 Aircraft Dynamics

With tools for modelling aircraft control surface forces, the longitudinal dynamics of the aircraft followed [27] and [33] to express the positional acceleration $\ddot{P}$ and rotational acceleration $\ddot{\theta}$ of the aircraft by:

$$m\ddot{P} = R(\theta - \alpha)F_w + R(\theta - \alpha)F_e + R(\theta + \xi)F_{fr} + R(\theta)F_{rr} + mg, \quad (3.9)$$

$$J_{yy}\ddot{\theta} = l_w R(-\alpha) + F_{w\perp} + (l \cos(\varepsilon) + l_e)R(-\alpha)F_{e\perp} - l_{fr}R(\theta + \xi)F_{fr\perp} + l_{rr}R(\theta)F_{rr\perp}, \quad (3.10)$$

Where m is the mass of the vehicle, and J_{yy} is the moment of inertia about the y -axis, θ is the pitch angle of the aircraft, and α defines the flight angle of attack. The terms l_w, l, l_e, l_{fr} , and l_{rr} describe the distances from the centre of mass of the aircraft to the centre of pressure of their respective control surfaces. The quantities F_w, F_e, F_{fr} , and F_{rr} represent the aerodynamic forces generated by the wings, elevator, front rotors, and rear rotors, respectively. The subscript $\perp$, denotes the force component that is orthogonal to the horizontal axis of the aircraft after rotation, with the clockwise rotation matrix R by angle $\angle$ defined as:

$$R(\angle) = \begin{bmatrix} \cos(\angle) & \sin(\angle) \\ -\sin(\angle) & \cos(\angle) \end{bmatrix}. \quad (3.11)$$

3.3 Optimal Control

The goal of optimal control is to find a way to control a dynamical system in such a way that certain criteria of optimality are satisfied. If the form of the function f is and $\mathbf{u}(t)$ is known over a time interval of interest, from an initial state, $\mathbf{Y}_0$, then equation 3.1 can be solved to find $\mathbf{Y}(t)$. It is then possible to determine a set of $\mathbf{u}(t)$ such that a cost function J is minimised over the time interval:

$$J = \int_0^T F[(\mathbf{Y}(t), \mathbf{u}(t), t)]dt + S[\mathbf{Y}(T), T] \quad (3.12)$$

In equation 3.12 F is a function that reflects the cost associated with changes in the state or input variables. While S is a final state cost ensuring that the final position of the system converges to the requirements[20]. The system can be restricted with specified state and input constraints to represent the desired dynamics, creating a powerful framework to specify complex behaviours with simple objective functions [30].

Optimal control is often separated into two major components, open-loop design and closed-loop control. The open-loop design refers to solving the cost function J and obtaining an optimality that being either a local minimum or maximum. However, in practice a feedback controller is often

required to ensure adherence of the system to the solution and robustness. This section will develop an understanding of the literature surrounding open-loop design and closed-loop control.

3.3.1 Trajectory Generation

There are various trajectory generation techniques for the open-loop design component of optimisation problems that are generally separated into two categories for determining optimal solutions: direct and indirect. Direct methods first discretise the problem, then incorporate numerical approximate solution techniques to iteratively improve the estimations until an optimality is found. However, an indirect method initially derives the Pontryagin conditions (necessary and sufficient) for optimality, then solves the resulting boundary value problem numerically. Although indirect approaches have shown greater accuracy in optimising cost functions, direct methods are more widely used as they are much easier to setup and show larger regions of convergence [9, 10]. As this problem involves a specific vehicle and task, it would be inefficient to introduce an indirect approach for trajectory generation. A direct method can take advantage of understood boundaries and constraints to computation and setup complexities.

Mathew Kelly (2017) develops a sound understanding of direct methods in *An Introduction to Trajectory Optimization* [18], stating that direct methods vary in complexity. although there is no specific best control method, the selection of the best approximation method depends on the problem itself. Single and multiple shooting methods are used when dynamics must be computed accurately, but the control trajectory is simple. Examples include spacecraft and other ballistic trajectories. Collocation methods are applicable to more problems that require accurate depictions of dynamics and control. These methods can be separated further depending on the order of the segment approximation. High-order collocation methods are known as orthogonal collocation, while trapezoidal collocation is understood as a low-order method and Hermite-Simpson a medium-order method. The selection of a collocation method is generally done by beginning with a low-order method and performing an error analysis to understand if the trajectory should be re-meshed or a higher-order approximation is required [18].

The collocation method refers to the approximation tool for estimating trajectories. The methods are essentially used to represent continuous-time problems as non-linear systems. This is done by choosing n time segments and discretising the problem so that $dt = T/n$. Given that at every sample node the problem has m states, the overall system now has $m \times n$ states. The collocation method is responsible for ensuring that the system dynamics are followed such that the next collocation point follows the previous via the system dynamics implemented as collocation constraints. This is iterated over the whole trajectory and refined so that a minimum or maximum is obtained depending on the problem. The degree of order refers to the degree of estimation. Low-order methods such as the trapezoidal method, use linear spline approximations, and have lower accuracy than higher-order methods such as the Hermite-Simpson method, which implements polynomial spines with greater accuracy at the cost of computation cost [18]. Given that the estimation converges to a solution, the solution will be a set of states and inputs at each collocation point. This is the open-loop solution to the cost function for the dynamical system. Between these points estimation tools are used depending on the approximation method used to interpolate states and inputs between the collocation points.

Matthew Kelly, who supplies much of the literature followed in this section for trajectory generation, also supplies a Matlab library for trajectory generation that will be highly useful for the project [19]. Following Kelly's recommendation, the trapezoidal method will initially be explored to reduce

computational costs, then higher-order methods will be implemented if an error analysis deems it necessary.

3.3.2 Closed-Loop Control

Closed-loop control is highly important for most systems, especially autonomous control, as it offers the possibility of dynamically adjusting inputs to achieve desired states and reduce the ill effects of disturbances during operation, increasing robustness and reliability [12, 26]. The controller considered for this project is the linear quadratic regulator (LQR). "The linear quadratic regulator is likely the most important and influential result in optimal control theory to date" [30]. This offline controller minimises a quadratic cost function of a linear dynamical system to determine the optimal control strategy while maintaining the integrity of the open-loop optimal design. The LQR is desirable as a controller as the offline computation allows for less energy expenditure during flight compared to online controllers and the simple and intuitive implementation allows for broad application. The use of LQR by Willis et al. (2020) in the *State-Dependent LQR Control for a Tilt-Rotor UAV* [32] paper shows a strong example of the use of the LQR controller in the adherence of aircraft trajectory and provides a basis for implementing the LQR in this project.

From Tedrake's (2023) *Underactuated Robotics*[30], a linear dynamical system can be represented by the first-order partial derivative matrices (Jacobian matrices) of the dynamics function A and B such that:

$$\dot{\mathbf{Y}}(t) = A(t) \mathbf{Y}(t) + B(t) \mathbf{u}(t) \quad (3.13)$$

where

$$A(t) = \frac{\partial f(t)}{\partial \mathbf{Y}(t)}, \quad B(t) = \frac{\partial f(t)}{\partial \mathbf{u}(t)}. \quad (3.14)$$

These matrices are highly useful, as they allow for the representation of nonlinear dynamical systems (in this case the aircraft's dynamics f) as linear dynamical systems around equilibrium points by small error approximation. Due to the open-loop trajectory design, at a given time, the project is able to represent the state and input error by the difference between measured state/input and the desired state/input specified by the open-loop solution:

$$\bar{\mathbf{Y}}(t) = \mathbf{Y}(t) - \mathbf{Y}_d(t), \quad \bar{\mathbf{u}}(t) = \mathbf{u}(t) - \mathbf{u}_d(t). \quad (3.15)$$

This allows for system equilibrium to be represented when the aircraft state and input equals the desired state and input ($\mathbf{Y}(t) = \mathbf{Y}_d(t)$ and $\mathbf{u}(t) = \mathbf{u}_d(t)$). At a given time, A and B can be developed by measuring the aircraft's dynamics gradient arising from small state and input perturbations from the open-loop solution. allowing the system to be linearly represented by:

$$\dot{\bar{\mathbf{Y}}}(t) = A(t) \bar{\mathbf{Y}}(t) + B(t) \bar{\mathbf{u}}(t). \quad (3.16)$$

Given a quadratic cost function defined by:

$$J = \int_{t_0}^{t_f} (\bar{\mathbf{Y}}(t)^T Q \bar{\mathbf{Y}}(t) + \bar{\mathbf{u}}(t)^T R \bar{\mathbf{u}}(t)) dt. \quad (3.17)$$

where $Q \in \mathbb{R}^{10 \times 10}$ is the diagonal state error cost matrix, and $R \in \mathbb{R}^{4 \times 4}$ is the diagonal input error cost matrix. The time varying feedback control that minimises the value of the cost is given by:

$$\bar{\mathbf{u}}(t) = K(t) \bar{\mathbf{Y}}(t). \quad (3.18)$$

Where the time varying gain matrix ($K(t)$) is given by:

$$K(t) = R^{-1}B^T(t)S(t). \quad (3.19)$$

and $S(t)$ is determined by solving the Riccati equation:

$$-\dot{S}(t) = A^T(t)S(t) + S(t)A(t) - S(t)B(t)R^{-1}B^T(t)S(t) + Q. \quad (3.20)$$

The Riccati equation can be solved by various methods, however as the trajectory requires the aircraft to finish in the requires state, the best method for this project is to integrate backwards from t_f to t_0 [27].

The cost function minimising input from Eq. 5.6 can then be applied to the system by:

$$\mathbf{u}(t) = \mathbf{u}_d(t) - K(t)\bar{\mathbf{x}}(t) \quad (3.21)$$

To implement this closed-loop control, Tedrake's book (2023) on *Underactuated Robotics* provides various optimal control methods. This project will utilise the chapter on LQR control and follow the documentation for Matlab implementation[30].

Aircraft Modelling

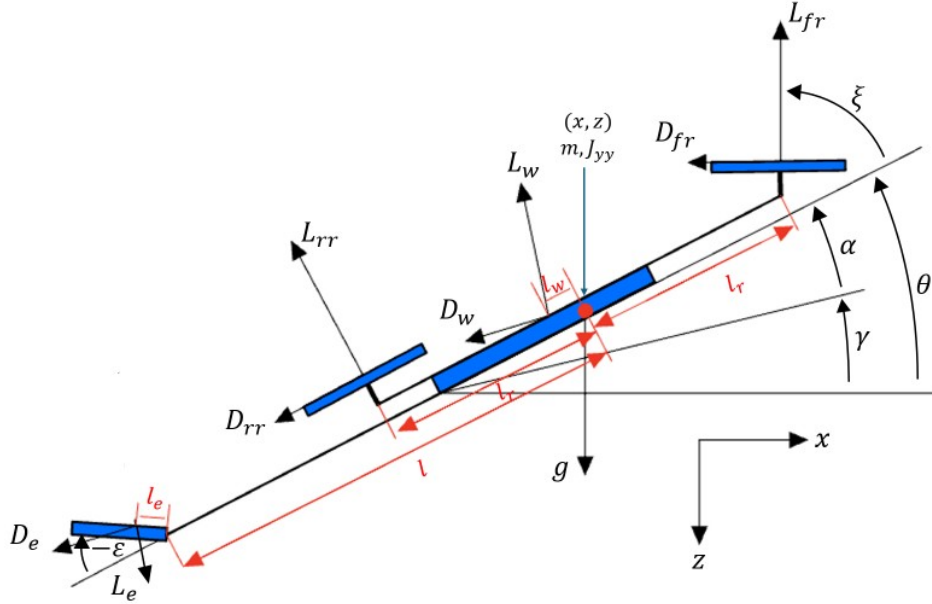


Figure 4.1: Griffin Pro aircraft model: The blue segments represent the aircraft's control surfaces, with areas S_w , S_e , and S_r respectively, going into the page. Aerodynamic and propulsive forces are modelled at the surface's centre of pressure. The red segments represent the distance between the aircraft's centre of mass and the centre of pressure of the respective control surface. For model parameter values and dimensions, see Table 4.1.

Following Section 3.2, the goal of this chapter is to model the complex nonlinear aerodynamics, rotor energy consumption and represent the Griffin Pro as a dynamical system, comprising ten states ($\mathbf{Y}(t) \in \mathbb{R}^{10}$) and four inputs ($\mathbf{u}(t) \in \mathbb{R}^4$) as given Table 2.2, such that:

$$\dot{\mathbf{Y}}(t) = f(\mathbf{Y}(t), \mathbf{u}(t), t),$$

where $f : \mathbb{R}^{10} \times \mathbb{R}^4 \rightarrow \mathbb{R}^{10}$ represents the longitudinal dynamics given by Equations 3.9 and 3.10. Restated below for convenience:

$$m\ddot{P} = R(\theta - \alpha)F_w + R(\theta - \alpha)F_e + R(\theta + \xi)F_{fr} + R(\theta)F_{rr} + mg, \quad (4.1)$$

$$J_{yy}\ddot{\theta} = l_w R(-\alpha) + F_{w\perp} + (l \cos(\varepsilon) + l_e)R(-\alpha)F_{e\perp} - l_r R(\theta + \xi)F_{fr\perp} + l_r R(\theta)F_{rr\perp}, \quad (4.2)$$

where $\ddot{P}$ represents the positional acceleration of the aircraft, $\ddot{\theta}$ denotes the rotational acceleration, m is the mass of the vehicle and J_{yy} is the moment of inertia about the y -axis. The variable θ corresponds to the pitch of the aircraft, while α defines the flight angle of attack. The terms l_w, l, l_e, l_{fr} , and l_{rr} describe the distances from the centre of mass of the aircraft to the centre of pressure of their respective control surfaces. The quantities F_w, F_e, F_{fr} , and F_{rr} represent the total forces generated by the wings, elevator, front rotors, and rear rotors, respectively. The subscript $\perp$, denotes the force component orthogonal to the horizontal axis of the aircraft after rotation. $R(\angle)$ represents the clockwise rotation matrix by angle $\angle$, defined in Equation 3.11.

The Griffin Pro consists of a traditional aeroplane design comprising a fuselage, wings, tail, and nose, with the addition of four rotors positioned around the aircraft's centre of mass. To determine the control surface forces in the mathematical dynamics model, the system assumes that the aircraft can be modelled as a rigid body subject to aerodynamic, propulsive, and gravitational forces in a low-altitude, windless environment[27]. The aircraft dynamics were separated into aeroplane dynamics, comprising aerodynamic forces developed in the wings and tail as functions of the respective control surface's velocity and angle of attack, and quadrotor dynamics, comprising propulsive and aerodynamic forces generated from the angular rate and airflow through the four rotors (see Figure 4.1).

The physical and environmental parameters used within the modelling and control sections are expressed in Table 4.1. Physical parameters were estimated from ARACE Griffin Pro design drawings and Ardupilot vehicle data. The environmental parameters follow those reported in [27].

The model follows the North-East-Down (NED) convention, standard for aerospace [5, 26, 32], restricted to the x and z axes (North-Down). The convention is defined with respect to the inertial frame, consistent with physical systems that track the vehicle via GPS coordinates. Note that the z axis is positive in the downward direction and was replaced by height in the graphs, according to $h = -z$, for a simpler interpretation.

The generation of a high-fidelity model representative of the Griffin Pro was paramount to the trajectory optimisation design and control problem; this chapter details the production of the model that was capable of fulfilling this task and meeting the modelling goals.

Parameter	Value	Parameter	Value
g	9.81 m/s ²	l	0.74 m
ρ	1.2682 kg/m ³	l_w	0.03 m
m	7 kg	l_e	0.04 m
J_{yy}	0.39725 kg-m ²	l_{fr}	0.31 m
S_w	0.45 m ²	l_{rr}	0.41 m
S_e	0.105 m ²	N_b	2
R_r	0.175 m	c	0.025 m

Table 4.1: Model parameters for the Griffin Pro.

4.1 Aeroplane Forces

The aeroplane model is described by the aerodynamic forces that the fixed-wing control surfaces experience in motion. The forces are represented by vectors relative to the centre of pressure of the respective control surface and are proportional to the local velocity and angle of attack. Hence, the local velocity experienced at the centre of pressure for each fixed-wing control surface (wing and tail)[27] was defined with respect to the inertial frame:

$$V_w = \begin{bmatrix} \dot{x} + l_w \dot{\theta} \sin \theta \\ \dot{z} + l_w \dot{\theta} \cos \theta \end{bmatrix}, \quad V_e = \begin{bmatrix} \dot{x} + l \dot{\theta} \sin \theta + l_e (\dot{\theta} + \dot{\varepsilon}) \sin(\theta + \varepsilon) \\ \dot{z} + l \dot{\theta} \cos \theta + l_e (\dot{\theta} + \dot{\varepsilon}) \cos(\theta + \varepsilon) \end{bmatrix}, \quad (4.3)$$

The angle of attack (α) was then defined as the angle between the orientation of the aircraft control surfaces and the oncoming airflow (Equations 4.4), including the control surface deflection angle. As this model does not include the effects of wind, the direction of oncoming airflow is anti-parallel to

the flight-path direction (γ) (computed using the **atan2()** function in MATLAB) [27]:

$$\alpha_w = \theta - \gamma + 4\frac{\pi}{180}, \quad \alpha_e = \theta - \gamma + \varepsilon, \quad (4.4)$$

where

$$\gamma = \tan^{-1}\left(\frac{-\dot{z}}{\dot{x}}\right). \quad (4.5)$$

From Equations 4.3 and 4.4, Equation 3.3 was used to express the aerodynamic forces developed on each control surface by:

$$L_w = \frac{1}{2}\rho|V_w|^2 S_w C_L(\alpha_w), \quad D_w = \frac{1}{2}\rho|V_w|^2 S_w C_D(\alpha_w), \quad (4.6)$$

$$L_e = \frac{1}{2}\rho|V_e|^2 S_e C_L(\alpha_e), \quad D_e = \frac{1}{2}\rho|V_e|^2 S_e C_D(\alpha_e). \quad (4.7)$$

Where ρ is the density of air, S_w and S_e represent the control surface areas of the wings and tail, respectively. The values of the parameters are shown in Table 4.1. C_L and C_D are dimensionless aerodynamic coefficient functions with respect to angle of attack, as developed in the next section. Drag is generated anti-parallel to the direction of aircraft motion, and lift is orthogonal to drag. According to established convention, the forces developed rightward or downward are positive, and those developed leftward or upward are negative. Note that the aerofoil in the elevator is orientated upside down, relative to the wing's aerofoil orientation, to improve the static stability of the airframe.

4.1.1 Lift and Drag Models

Developing Griffin Pro-specific models for lift and drag coefficients was achieved by blending the linear region of the Clark Y aerofoil data, used in the Griffin Pro's wing and tail design, with Flat Plate Theory. Using a Reynolds number of 200,000 and a Ncrit value of 9, the data for the Clark Y aerofoil were obtained from Airfoilttools.com [2]. The data for each coefficient were linearly approximated by:

$$C_{L_{lin}}(\alpha) = C_{L_0} + C_{L_\alpha}\alpha, \quad \alpha \in [-5.5^\circ, 10^\circ], \quad (4.8)$$

$$C_{D_{lin}} = C_{D_0} + C_{D_\alpha}\alpha, \quad \alpha \in [-0.5^\circ, 10^\circ]. \quad (4.9)$$

Where C_{L_0} and C_{D_0} are the zero angle of attack lift and drag coefficients, respectively, and C_{L_α} and C_{D_α} are the lift and drag gradients with respect to α in radians, respectively. These coefficient values are presented in Table 4.2. Linearised aerofoil data can then be blended with Flat Plate Theory (Equations 3.2) using a sigmoid function [8, 33]. The lift and drag coefficient functions for Equations 4.6 and 4.7 are expressed as:

$$C_L(\alpha) = (1 - \sigma(\alpha))C_{L_{lin}}(\alpha) + \sigma(\alpha)C_{L_{fp}}(\alpha), \quad (4.10)$$

$$C_D(\alpha) = (1 - \sigma(\alpha))C_{D_{lin}}(\alpha) + \sigma(\alpha)C_{D_{fp}}(\alpha), \quad (4.11)$$

where

$$\sigma(\alpha) = \frac{1 + e^{-M(\alpha-rb)} + e^{M(\alpha-lb)}}{(1 + e^{-M(\alpha-rb)})(1 + e^{M(\alpha-lb)})}. \quad (4.12)$$

The resulting aerodynamic coefficient functions are shown in Figures 4.2 and 4.3. It should be noted that the sigmoid function differed for lift and drag, depending on the parameters M, rb, and lb input to the function, according to the values listed in Table 4.2.

Lift Coefficient	Value	Drag Coefficient	Value
C_{L_0}	0.4424	C_{d_0}	0.01
C_{L_α}	5.8151	C_{D_α}	0.05
M_{cl}	1	M_{cd}	26
rb_{cl}	0.28	rb_{cd}	0.28
lb_{cl}	-0.18	lb_{cd}	-0.05

Table 4.2: Parameter values used in modelling aeroplane control surface lift and drag coefficients.

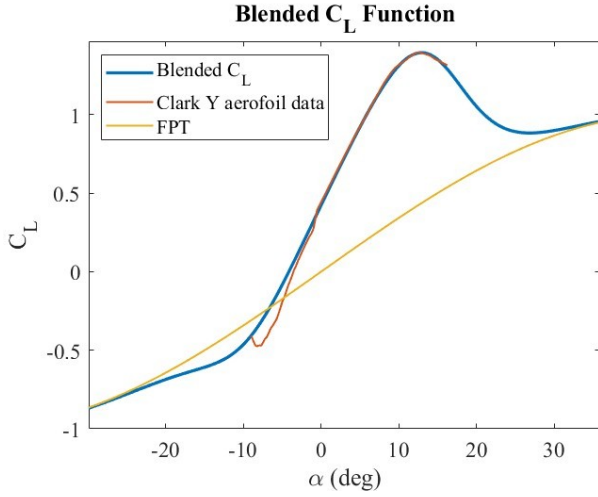


Figure 4.2: Blended Clark Y aerofoil data and Flat Plate Theory for coefficient of lift function w.r.t. angle-of-attack

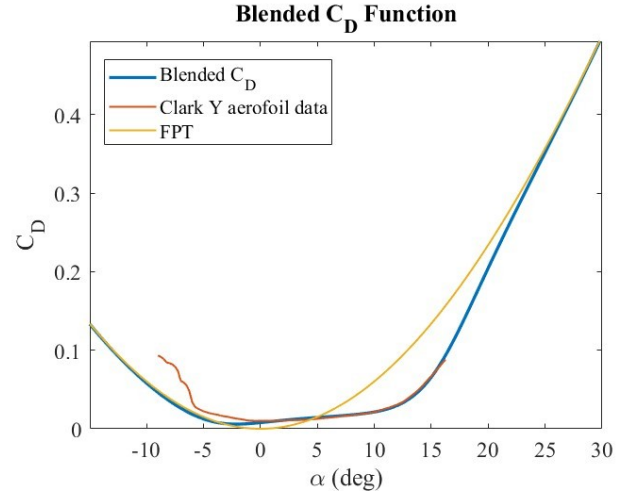


Figure 4.3: Blended Clark Y aerofoil data and Flat Plate Theory for coefficient of drag function w.r.t. angle-of-attack

A strong representation of the Clark Y aerofoil during positive angle of attack flight (as seen in Figures 4.2 and 4.3) was highly important for the fidelity of the model, as positive angle of attack flight was expected to occur throughout the landing trajectory. Although the functions are less representative of the aerofoil during negative angle of attack flight, since this flight regime is less likely, the representation following Flat Plate Theory still allows the control to transition the aircraft through the landing phase without destabilising the system.

4.2 Quadrotor Forces

To develop a representative model of Griffin Pro, the aircraft dynamics required the inclusion of the induced lift and drag forces developed in the rotors. As the modelling was restricted to the longitudinal frame, in this two-dimensional analysis, the front and rear rotors were modelled in pairs. In addition, it was assumed that within a rotor pair, the forces developed are symmetric between the two individual rotors, allowing the force developed in a pair to be expressed as twice the force generated by a single rotor.

Following Equations 3.4 and 3.5, the lift and drag forces developed in the front and rear rotor pairs were estimated with respect to the rotor speed by:

$$L_{fr} = \frac{1}{2} N_b \rho c R_r^3 \bar{\omega}_{fr}^2 C_T, \quad D_{fr} = \frac{L_{fr} |V_{h_{fr}}|}{R_r \bar{\omega}_{fr}}, \quad (4.13)$$

$$L_{rr} = \frac{1}{2} N_b \rho c R_r^3 \bar{\omega}_{rr}^2 C_T, \quad D_{rr} = \frac{L_{rr} |V_{hrr}|}{R_r \bar{\omega}_{rr}}. \quad (4.14)$$

Where N_b is the number of blades on a rotor, c is the blade chord (an assumed constant) and R_r is the rotational radius of the rotor, as expressed in Table 4.1. C_T is the thrust coefficient of the rotor, and V_h refers to the vehicle speed parallel to the respective rotor blade axis, defined as:

$$\begin{bmatrix} V_{hfr} \\ V_{vfr} \end{bmatrix} = \begin{bmatrix} \cos(-\Xi) & \sin(-\Xi) \\ -\sin(-\Xi) & \cos(-\Xi) \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix}, \quad \begin{bmatrix} V_{hrr} \\ V_{vrr} \end{bmatrix} = \begin{bmatrix} \cos(-\theta) & \sin(-\theta) \\ -\sin(-\theta) & \cos(-\theta) \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{z} \end{bmatrix}, \quad (4.15)$$

where

$$\Xi = \theta + \xi - \frac{\pi}{2},$$

and V_v denotes the vehicle speed orthogonal to the respective rotor blade axis. The thrust coefficient is variable depending on the pitch, advanced ratio, and air inflow rate of the rotor [7, p. 21]. However, it was estimated when the vehicle is in hover by balancing forces, when the lift and rotor speed are equal in all rotors, and counteract gravitational force, i.e.:

$$\begin{aligned} mg &= L_{fr} + L_{rr} \\ &= 2L_{fr} \\ &= N_b \rho c R_r^3 \bar{\omega}_{fr}^2 C_T. \end{aligned}$$

The equation was arranged for C_T as:

$$C_T = \frac{mg}{N_b \rho c R_r^3 \bar{\omega}_{fr}^2}. \quad (4.16)$$

Using model parameters (Table 4.1) and assuming that the Griffin Pro hovers at individual rotor speeds of 5500 RPM [7, p. 18], C_T was calculated. The thrust coefficient was then implemented as a constant with this calculated value, allowing the rotor thrust and drag to be estimated.

4.2.1 Rotor Power

The trajectory optimisation problem seeks to reduce the energy consumption of the landing transition phase, hence a method for calculating the rotor power was required. With rotor thrust and drag models now available, momentum theory was incorporated to estimate the power expended by each rotor. The momentum theory equations [7] for the lift, drag, and power developed in one rotor are defined as:

$$L_r = 2\rho\pi R_r^2 v_z^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (4.17)$$

$$D_r = 2\rho\pi R_r^2 v_x^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (4.18)$$

$$P_r = L_r(v_z^i - V_{vr}) + D_r(v_x^i + V_{hr}), \quad (4.19)$$

where v_z^i and v_x^i are the rotor-induced airflow components orthogonal and parallel to the rotor blades, respectively, and V_{vr} was derived as in Equation 4.15. Using definitions for L_r and D_r , from Equations 3.4, v_z^i , v_x^i and P_r were expressed[1] as:

$$(1 + \mu_r^2)v_z^{i4} + 2(\mu_r V_{hr} - V_{vr})v_z^{i3} + (V_{hr}^2 + V_{vr}^2)v_z^{i2} - K^2\bar{\omega}_r^4 = 0, \quad (4.20)$$

$$v_x^i = \mu_r v_z^i, \quad (4.21)$$

$$P_r = \Lambda\bar{\omega}_r^2((1 + \mu_r^2)v_z^i + \mu_r V_{hr} - V_{vr}), \quad (4.22)$$

where

$$K = \frac{N_b c R_r C_T}{8\pi}, \quad \Lambda = \frac{1}{4} N_b \rho c R_r^3 C_T.$$

The complete derivation [1] is presented in Appendix B. The quartic was solved using the MATLAB **roots()** function. Then, taking the maximum real root for v_z^i , the system was solved for P_r . This allowed the system to estimate the power produced by the rotor lift and drag forces at any given time, enabling the optimal trajectory design to minimise rotor energy consumption. The assumption still held that the power supplied to a rotor pair is double the power supplied to a single rotor. This power derivation does not account for motor inefficiencies, with expected losses of approximately 30% [7, p. 9], inferring that the total power consumption of the vehicle is greater than that expended as rotor lift and drag.

4.3 Aircraft Dynamics

The previous sections allow the Griffin Pro to be depicted as a rigid body subject to aerodynamic, propulsive, and gravitational forces, as illustrated in Figure 4.1. To implement the lift and drag forces in the longitudinal dynamics of the aircraft (Equations 4.1 and 4.2), these dynamics were expanded from the total force into a lift and drag representation:

$$\begin{bmatrix} m\ddot{x} \\ m\ddot{z} \end{bmatrix} = R(\theta)R(-\alpha) \begin{bmatrix} -D_w \\ -L_w \end{bmatrix} + R(\theta)R(-\alpha) \begin{bmatrix} -D_e \\ L_e \end{bmatrix} + R(\theta)R(\xi) \begin{bmatrix} T_{fr} \\ -D_{fr} \end{bmatrix} + R(\theta) \begin{bmatrix} -D_{rr} \\ -T_{rr} \end{bmatrix} + \begin{bmatrix} 0 \\ mg \end{bmatrix}, \quad (4.23)$$

$$\begin{bmatrix} * \\ J_{yy}\tau \end{bmatrix} = l_w R(-\alpha) \begin{bmatrix} -D_w \\ -L_w \end{bmatrix} + (l \cos(\varepsilon) + l_e) R(-\alpha) \begin{bmatrix} -D_e \\ L_e \end{bmatrix} - l_{fr} R(\xi) \begin{bmatrix} T_{fr} \\ -D_{fr} \end{bmatrix} + l_{rr} \begin{bmatrix} -D_{rr} \\ -T_{rr} \end{bmatrix}, \quad (4.24)$$

where

$$R(\angle) = \begin{bmatrix} \cos(\angle) & \sin(\angle) \\ -\sin(\angle) & \cos(\angle) \end{bmatrix}.$$

These dynamics were implemented in MATLAB and Simulink, following the methodology in [8], providing the necessary longitudinal dynamics model of the Griffin Pro UAV. This model met the objectives of modelling the complex aerodynamics of the Griffin Pro and the rotor power produced and formed the model foundation for the optimal landing transition control problem presented in the following chapter.

Optimal Control

The optimal control implementation, detailed in this chapter, was divided into the two associated project objectives: design an optimal trajectory that reduced the energy requirements of the landing system and ensures the safety of the aircraft and design an LQR controller that ensures the convergence of the system to the desired trajectory. The trajectory generation targeted the trajectory specific goals presented in Section 2.2.4 and implemented the aircraft safety constraints developed in Section 2.2.3. While, the LQR controller incorporated the requirements developed in Section 2.2.5.

5.1 Trajectory Generation

To design the trajectory optimisation problem that met the project re

To design the trajectory optimisation problem, the project used Matthew P Kelly’s MATLAB library[19]. The library implements its own **optimTraj()** function, which calls MATLAB’s **fmincon()** function to solve the underlying nonlinear program by invoking a direct collocation method that discretises the optimal control problem and numerically solves it as a constrained optimisation problem.

The OptimTraj User’s Guide provides details on interfacing with the library, explaining:

“The single calling sequence for using OptimTraj: **soln=optimTraj(problem)**, where the input **problem** is a struct that describes a trajectory optimisation problem, and the output **soln** is a struct that gives details regarding the solution to the trajectory optimisation problem” [22].

Here, ‘struct’ refers to a MATLAB data type. The input struct **problem** was defined by four primary fields that were modified to specify the trajectory optimisation problem, namely: **problem.func**, **problem.bounds**, **problem.options**, and **problem.guess**.

5.1.1 System Constraints

The field that handled the system constraints was **problem.func**. Within this field, two subfields were modified for the purpose of the project. The dynamics function that represents the longitudinal dynamics of the Griffin Pro was input to the field by:

$$\mathbf{problem.func.dynamics} : \dot{\mathbf{Y}}(t) = f(\mathbf{Y}(t), \mathbf{u}(t), t),$$

where $f : \mathbb{R}^{10} \times \mathbb{R}^4 \rightarrow \mathbb{R}^{10}$; the state of the system $\mathbf{Y}(t) \in \mathbb{R}^{10}$, and the input of the system $\mathbf{u}(t) \in \mathbb{R}^4$, were defined as developed in the previous chapter following the definitions in Table 2.2. These dynamics were then implemented as constraints that allow the solver to map states based on previous states and inputs as defined by the dynamics.

The **problem.func.pathObj** subfield was given a function that calculates the power produced by rotor lift and drag, according to Section 4.2.1, allowing the optimisation solver to minimise the rotor power consumed throughout the landing phase trajectory:

$$\mathbf{problem.func.pathObj} : J_P = P(\mathbf{Y}(t), t).$$

5.1.2 State and Input Constraints

All state and input constraints were defined in **problem.bounds**. This structure allowed the project to define the landing transition problem as developed in Section 2.2. The goal of reducing the transition phase to 7s was implemented in the bounds by:

$$\mathbf{problem.bounds.intialTime} : t_0 = 0,$$

and

$$\mathbf{problem.bounds.finalTime} : t_f = 7.$$

The final state of the trajectory required the aircraft to reach the origin and hover state, i.e., $\mathbf{Y}(t_f) = [0, 0, 0, 0, 0, 0, 4782.5, 5500, \pi/2, 0]^T$. It was experimentally determined that requiring the rotors to reach the exact hover-state speed resulted in excessive computational time. As the existing Ardupilot VTOL system stabilises the aircraft after the handover, this requirement was deemed unnecessary, provided the rotor speeds were close to the hover-state values. This was implemented in the **problem** struct by allowing the rotor speeds to reach a range near the hover state:

$$\mathbf{problem.bounds.finalState} = \mathbf{Y}(t_f) \pm [0, 0, 0, 0, 0, 0, 3000, 3000, 0, 0]^T.$$

To represent the cruise conditions experienced by the aircraft at Waypoint 12 and define aircraft states and inputs at the beginning of the trajectory, a trim analysis was conducted on the aircraft dynamics. The analysis solved the system for balanced flight at the specified speed and flight direction - in this case, 20m/s horizontal flight (flight angle of 0°) - such that the system could be represented by:

$$f(\mathbf{Y}^*, \mathbf{u}^*) = 0, \tag{5.1}$$

where the longitudinal dynamics of the aircraft are in equilibrium in state $\mathbf{Y}^*$, and input $\mathbf{u}^*$ [8]. The trim material from [8, pp. 65-67] was modified to accommodate the Griffin Pro model, and the MATLAB **fmincon()** function was used to minimise the difference between the desired trim states and the solution states, subject to constraints that match the states responsible for the vehicle's velocity with the desired speed and flight angle.

The resulting solution was $\mathbf{Y}^* = [*, *, 0.083, 20, 0, 0, 1355.34, 0.05, 0.087, -0.072]^T$ with zero input, i.e., $\mathbf{u}^* = [0, 0, 0, 0]^T$, noting that the longitudinal position is irrelevant to the trim state. This irrelevance allowed the problem design to implement the trajectory goals that blend the transition phase and VTOL descent and reduce the distance and duration of both by positioning the initial state of the Griffin Pro 50m horizontally and 20m vertically above the desired hover position:

$$\mathbf{problem.bounds.intialState} = [-50, -20, 0.083, 20, 0, 0, 1355.34, 0.05, 0.087, -0.072]^T.$$

The final constraint field related to the bounds on states and inputs. These constraints were implemented for three main reasons: to replicate the physical restrictions of the aircraft, to ensure its safety,

and to prevent actuator saturation. The bound implemented followed those developed in Section 2.2.3. Those constraints are then given to **problem** by:

$$\mathbf{problem.bounds.state.low} = [-\infty, -\infty, \frac{-\pi}{4}, -\infty, -\infty, \frac{-\pi}{4}, 0, 0, 0, \frac{-\pi}{4}]^T,$$

$$\mathbf{problem.bounds.state.upp} = [\infty, \infty, \frac{\pi}{4}, \infty, \infty, \frac{\pi}{4}, 10000, 10000, \frac{\pi}{2}, \frac{\pi}{4}]^T,$$

$$\mathbf{problem.bounds.control.low} = [-5000, -5000, \frac{-\pi}{4}, \frac{-\pi}{4}]^T,$$

and

$$\mathbf{problem.bounds.control.upp} = [5000, 5000, \frac{\pi}{4}, \frac{\pi}{4}]^T.$$

5.1.3 Collocation Method

The **problem.options** handles input options for both **optimTraj()** and **fmincon()** to solve the problem. Primarily, the collocation method to discretise and approximate the continuous trajectory optimisation problem by approximating it as a sequence of segments [18] is specified here. The project used the low-order trapezoidal collocation method:

$$\mathbf{problem.options.method} = 'trapezoid'.$$

“The trapezoid method works by making the assumption that the dynamics, objective function, and control trajectories can be approximated with a linear spline, and the state trajectory is a quadratic spline, obtained by integrating the dynamics’ linear spline” [22].

” This method was selected over higher-order methods, such as the Hermite-Simpson method, because of lower computational requirements and compatibility with linear interpolation, which simplified both offline and online components of the control design.

The trapezoid method employs **s** segments:

$$\mathbf{problem.options.trapezoid.nGrid} = \mathbf{s}.$$

Generally, the greater the number of segments, the greater the approximation accuracy of the trapezoid method; however, increasing the number of segments also increased computational time. Segments of 0.05s in length (**s** = 141 segments for a 7s trajectory) were deemed sufficiently accurate to approximate the dynamics, while taking approximately 24 for the system to solve; fewer segments caused the accuracy to drop and any more resulted in excessive computational time.

5.1.4 Initial Guess

The **optimTraj()** requires an initial guess, which the function interpolates to evaluate values on the solution grid. When running trajectory optimisation problems with a lower number of segments, the solutions were generated quickly, allowing the input problem to be evaluated and adjusted if necessary, until a sufficient trajectory was determined.

The solution state and inputs were then entered as an accurate initial guess for a problem with a higher number of segments, reducing the computation time. This technique allowed for fast adjustments and reduced the computation time for 141-segment problems to approximately 14 hours when a strong initial guess was provided.

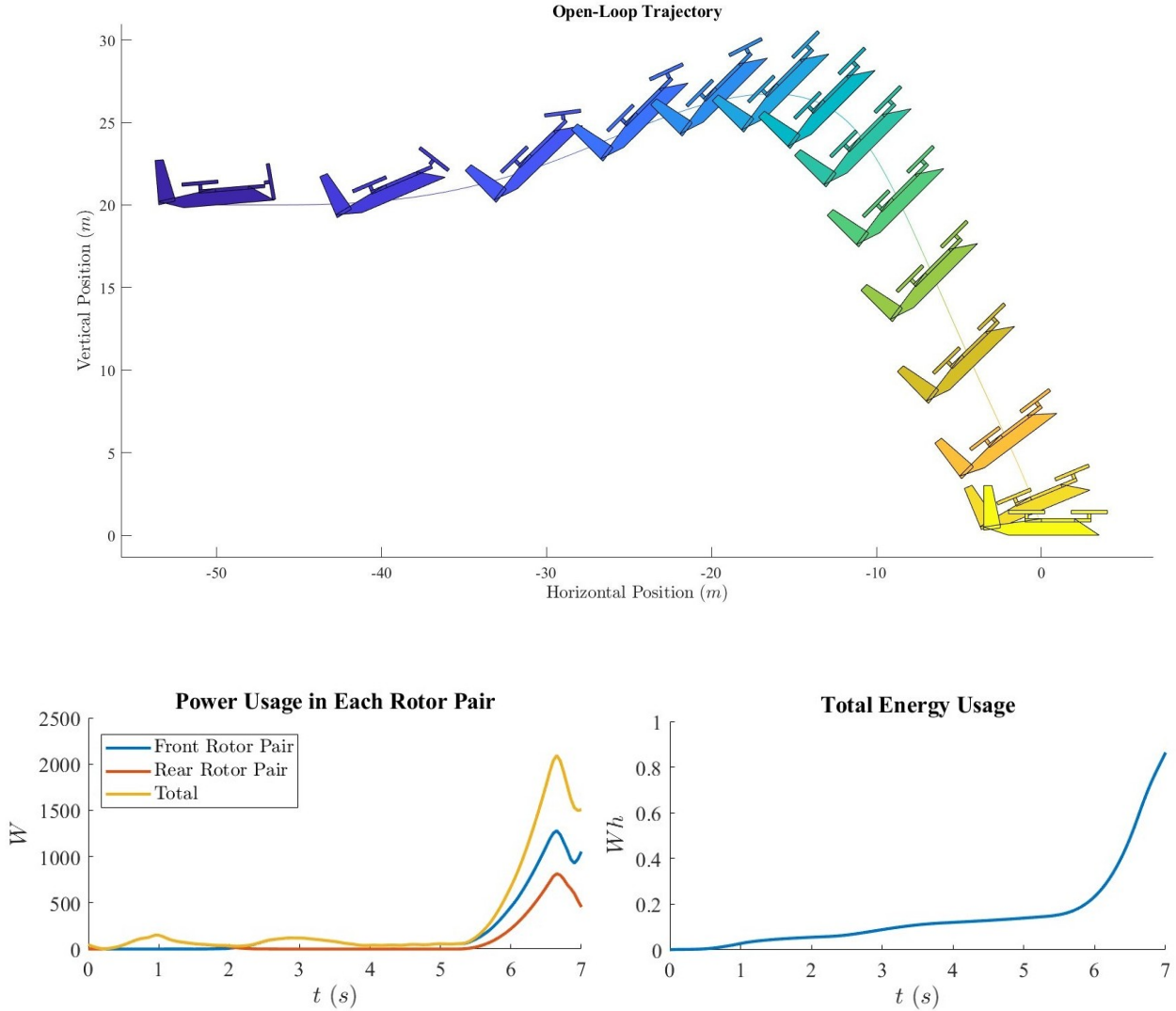


Figure 5.1: Visualisation of the trajectory, power and energy usage determined in the solution of the trajectory optimisation problem stated in Section 5.1.

5.1.5 The Trajectory Optimisation Solution

With the trajectory optimisation problem defined, the project objective of reducing the cost function was given as [22]:

$$\min_{\mathbf{Y}(t), \mathbf{u}(t)} \int_{t_0}^{t_f} P(\mathbf{Y}(\tau), \tau) d\tau,$$

Where P is the power estimation function, subject to the constraints:

$$\dot{\mathbf{Y}}(t) = f(\mathbf{Y}(t), \mathbf{u}(t), t),$$

$$[-\infty, -\infty, \frac{-\pi}{4}, -\infty, -\infty, \frac{-\pi}{4}, 0, 0, 0, \frac{-\pi}{4}]^T \leq \mathbf{Y}(t) \leq [\infty, \infty, \frac{\pi}{4}, \infty, \infty, \frac{\pi}{4}, 10000, 10000, \frac{\pi}{2}, \frac{\pi}{4}]^T,$$

$$[-5000, -5000, \frac{-\pi}{4}, \frac{-\pi}{4}]^T \leq \mathbf{u}(t) \leq [5000, 5000, \frac{\pi}{4}, \frac{\pi}{4}]^T,$$

$$\mathbf{Y}(t_0) = [-50, -20, 0.083, 20, 0, 0, 1355.34, 0.05, 0.087, -0.072]^T,$$

$$[0, 0, 0, 0, 0, 0, 1782.5, 2500, \pi/2, 0]^T \leq \mathbf{Y}(t_f) \leq [0, 0, 0, 0, 0, 0, 7782.5, 8500, \pi/2, 0]^T.$$

Provided that the trajectory optimiser converged to a feasible solution, the **optimTraj()** function returned the struct **soln**. This struct contained the state and input values of the solution trajectory

(Figure 5.2) at grid points determined by the segments lengths, the value of the objective function, and the information for the trajectory optimisation solution.

The solution to the trajectory optimisation problem specified above, with 141 segments, is visualised in Figures 5.1 and 5.2. The figures depict the ability of the trajectory to minimise rotor usage and, subsequently, the power consumed by the system. The power developed purely as rotor lift and drag is estimated at $\approx 0.86Wh$ when incorporating the 67% efficiency factor ($0.58Wh \times 1.5$), over the trajectory.

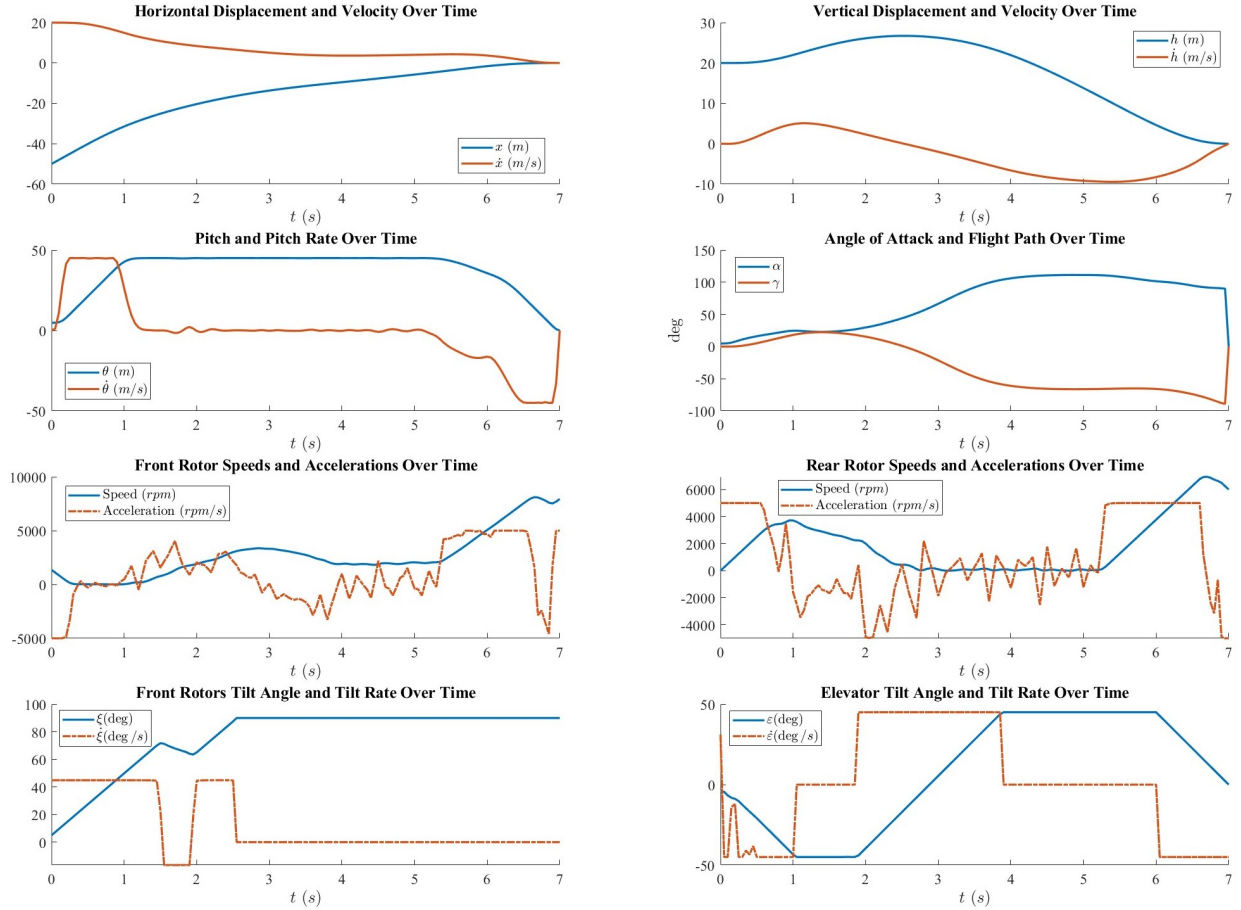


Figure 5.2: Griffin Pro states and inputs determined by the solution to the trajectory optimisation problem developed in Section 5.1. The states are represented by solid lines, and inputs are represented by dashed lines

5.1.6 Open-Loop Simulation

When the solution to the trajectory optimisation problem was simulated, as in Figure 5.3, we saw the system failing to follow the desired states (dashed lines) when using only the open-loop input control. Without the feedback gain, disturbances arise due to the dynamics approximation error from state and input interpolation and system stiffness, which caused the system to destabilise. Therefore, a feedback gain was required to dynamically adjust the input and ensure the convergence of the system.

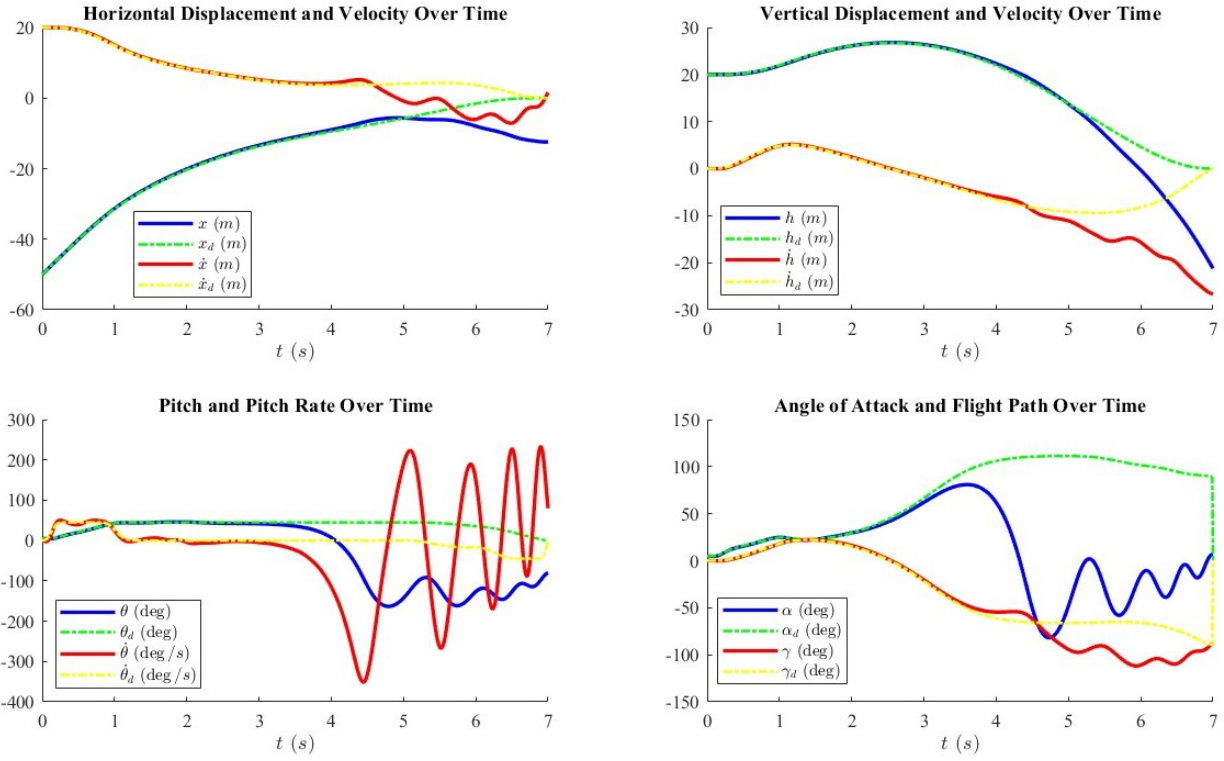


Figure 5.3: States vs desired states of the aircraft when the solution for the trajectory optimisation problem is simulated without feedback gain

5.2 The Linear-Quadratic Regulator Feedback Controller

Closed-loop control is highly important for most systems, especially in autonomous applications, as it offers the possibility of dynamically adjusting inputs to achieve desired states and reduce the adverse effects of disturbances during operation, thus increasing robustness and reliability [12, 26].

The linear-quadratic regulator (LQR) was selected for the closed-loop control because the feedback gain matrix was determined by solving the controller's quadratic cost function which aligned with the project goal of reducing energy consumption and would ensure the integrity of the landing phase trajectory designed in the previous section by minimising any additional energy usage by the controller while increasing the robustness of the system. Moreover, the gain matrix is computed offline, further ensuring that the energy required by the controller during flight is minimised.

To implement the controller, as depicted in the block diagram in Figure 5.4, the project followed the methodology and work presented in [32] and [30].

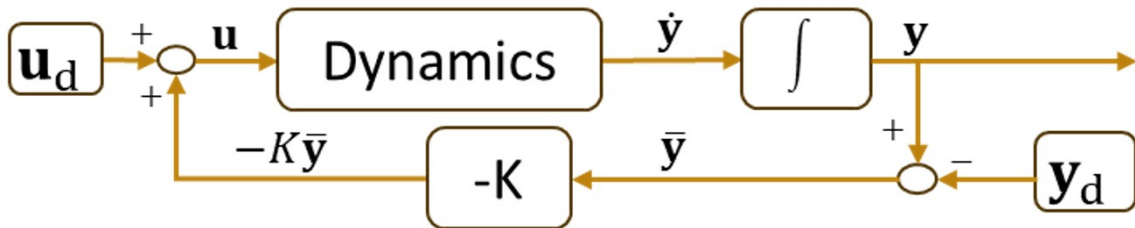


Figure 5.4: The block diagram of the feedback gain implementation of the LQR

5.2.1 Error-State Representation

With the desired trajectory determined according to Section 5.1, given that $\mathbf{Y}_d \in \mathbb{R}^{10 \times 141}$ and $\mathbf{u}_d \in \mathbb{R}^{4 \times 141}$ represent the states and inputs of the solution trajectory, by linearly interpolating between the grid point such that $\mathbf{Y}_d(t) \in \mathbb{R}^{10}$ and $\mathbf{u}_d(t) \in \mathbb{R}^4$, the error-state and input of the system were represented by:

$$\bar{\mathbf{Y}}(t) = \mathbf{Y}(t) - \mathbf{Y}_d(t), \quad \bar{\mathbf{u}}(t) = \mathbf{u}(t) - \mathbf{u}_d(t). \quad (5.2)$$

This suggested the error-state representation of the nonlinear aircraft dynamics:

$$\dot{\bar{\mathbf{Y}}}(t) = \bar{f}(\bar{\mathbf{Y}}(t), \bar{\mathbf{u}}(t), t). \quad (5.3)$$

Here, $\bar{f} : \mathbb{R}^{10} \times \mathbb{R}^4 \rightarrow \mathbb{R}^{10}$ represents the nonlinear dynamics that arises from disturbances from the desired trajectory. To implement the LQR, the controller required linearised dynamics. Using error-state representation, the nonlinear dynamics of the aircraft was linearly approximated by the Jacobian first-order partial derivative matrices A and B , such that [32]:

$$\dot{\bar{\mathbf{Y}}}(t) = A(t)\bar{\mathbf{Y}}(t) + B(t)\bar{\mathbf{u}}(t), \quad (5.4)$$

where

$$A(t) = \frac{\partial f(t)}{\partial \bar{\mathbf{Y}}(t)}, \quad B(t) = \frac{\partial f(t)}{\partial \bar{\mathbf{u}}(t)}. \quad (5.5)$$

The Jacobian matrices were developed by analysing the gradient arising in the system's dynamics when iteratively applying small perturbation in each of the system's states and inputs. To reduce the need for online interpolation, Jacobian matrices were constructed at every $1 \times 10^{-4}s$ time step by linearly interpolating the solution grid over the entire $7s$ trajectory. This time step was deemed small enough to prevent online interpolation during Simulink simulations, and rounding to the nearest timestamp, if necessary, would not compromise the integrity of the system.

5.2.2 Feedback Gain

The feedback gain used to increase the robustness of the system is expressed by:

$$\bar{\mathbf{u}}(t) = -K(t)\bar{\mathbf{Y}}(t). \quad (5.6)$$

To determine this gain, the LQR controller minimised a quadratic cost function such that the solution was the gain matrix [30]. The LQR cost function is defined by:

$$J = \int_{t_0}^{t_f} (\bar{\mathbf{Y}}(t)^T Q \bar{\mathbf{Y}}(t) + \bar{\mathbf{u}}(t)^T R \bar{\mathbf{u}}(t)) dt, \quad (5.7)$$

where $Q \in \mathbb{R}^{10 \times 10}$ is the diagonal state error cost matrix and $R \in \mathbb{R}^{4 \times 4}$ is the diagonal input error cost matrix. As can be seen in Figures 5.1 and 5.2, there is a strong relationship between rotor speed and rotor power use. This is expected due to the equations used to model the rotor power in Section 4.2.1.

To ensure the integrity of the solution trajectory and align with the controller objectives developed in Section 2.2.5, Q and R were experimentally design to ensure that the system reached a hover state, while minimising the rotor usage and, subsequently, the power consumed. The cost matrices are given as:

$$Q = \text{diag}(0.05, 1, 1000, 5, 10, 5000, 5 \times 10^{-7}, 5 \times 10^{-7}, 1 \times 10^4, 1),$$

$$R = \text{diag}(1 \times 10^{-6}, 1 \times 10^{-6}, 1, 100).$$

It must be noted that the error-states and inputs are measured in different units than can magnitudes in difference for a given state error value. For example, the pitch angle error ($\bar{\theta}$) is measured in radians, while the front rotor speed error ($\bar{\omega}_{fr}$) is measured in RPM, which means that to implement a relatively similar error cost, the scalar value must be magnitudes in difference. A primary desire of the controller design was to ensure that the system reached hover, allowing for safe handover should the system be disturbed. As such, large costs were placed on $\bar{\theta}$ and the pitch rate error ($\dot{\bar{\theta}}$) prioritising the safety of the aircraft. Accordingly, a high cost (essentially a controller constraint) was placed on the error in the tilt angle of the front rotors, to ensure that the actuator was not pushed beyond its boundaries of $\xi \in [0, \frac{\pi}{2}]$. Should the aircraft be disturbed during landing, low cost was placed on the horizontal position error ($\bar{x}$), as these landings occur in open spaces, any positional error was deemed not important enough to potentially jeopardise the safety of the aircraft by potentially destabilising the system in attempt to converge the position. A slightly higher cost was placed on the height error ($\bar{z}$) as VTOL descent is a major energy cost of the landing transition. Beyond this, the closed-loop controller was designed to utilise the input and state controls (rotors and elevator) to converge the system to the desired path. The associated values in the cost function were deemed sufficient to allow the feedback gain to dynamically adjust control while ensuring power consumption is minimised.

With the cost functions defined, the next step in determining the feedback gain was to compute $S(t)$ by solving the continuous-time Riccati equation [27, 30]:

$$-\dot{S}(t) = A^T(t)S(t) + S(t)A(t) - S(t)B(t)R^{-1}B^T(t)S(t) + Q. \quad (5.8)$$

The equation was solved numerically via Euler integration. Beginning at t_f and using a $1 \times 10^{-4}s$ time step, the trajectory was integrated back to t_0 , with a correction iteration at each time step to improve the integration estimate. With $S(t)$ determined, the time varying gain matrix $K(t)$ is given by:

$$K(t) = R^{-1}B^T(t)S(t). \quad (5.9)$$

This allows the LQR feedback to be implemented as in Figure 5.4 using Equations 5.2 and 5.6 such that:

$$\mathbf{u}(t) = \mathbf{u}_d(t) - K(t)\bar{\mathbf{x}}(t) \quad (5.10)$$

5.2.3 Successful Feedback Control

The trajectory system was then simulated with the feedback gain applied (Figures 5.5 and 5.6). When starting in the desired initial state, we see a significant improvement in the system's ability to adhere to the desired trajectory compared to the state graphs depicted in Figure 5.3. In Figure 5.6, the system dynamically adjusts the rotor speeds and elevator angle to converge the system to the path over the trajectory. In Figure 5.7, we observe the controller's ability to maintain low power usage, thereby ensuring the integrity of the trajectory design. However, a true test of any feedback controller is its ability to handle disturbances.

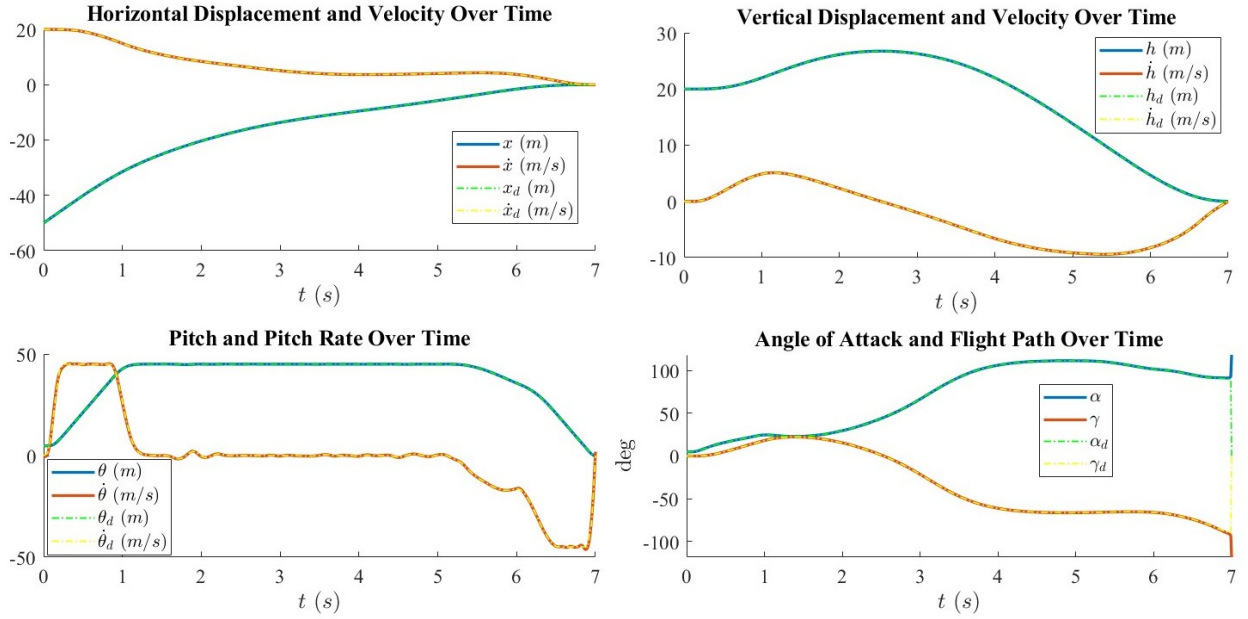


Figure 5.5: States vs desired states when feedback gain is applied to the system with an LQR controller

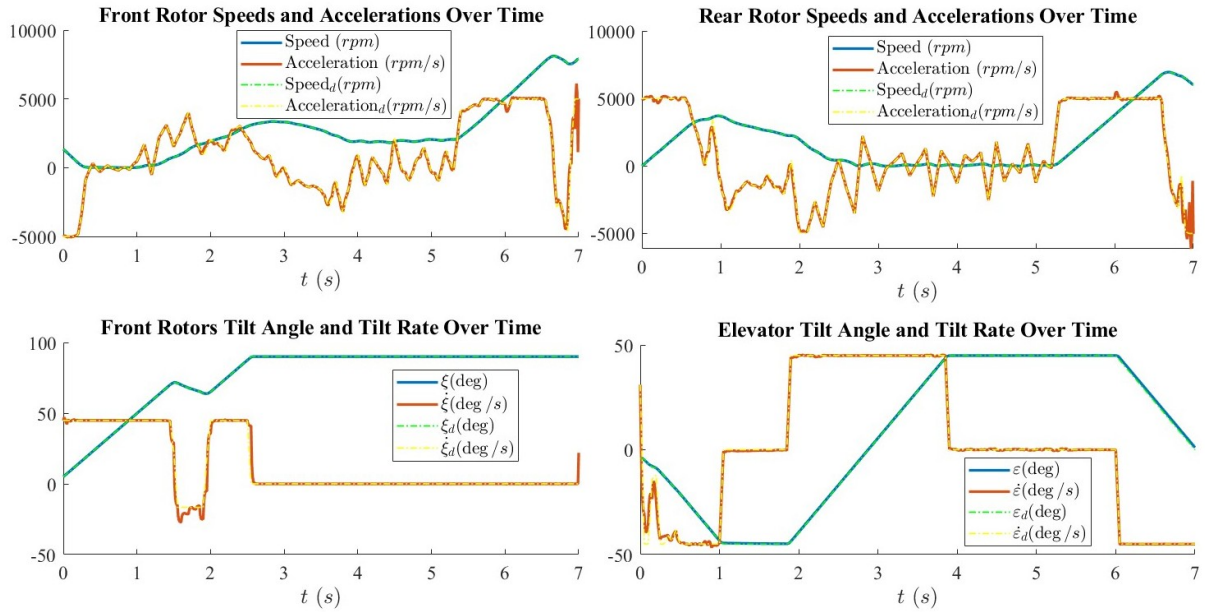


Figure 5.6: Inputs vs desired inputs when feedback gain is applied to the system with an LQR controller. The controller is dynamically adjusting inputs to converge the system to the desired path

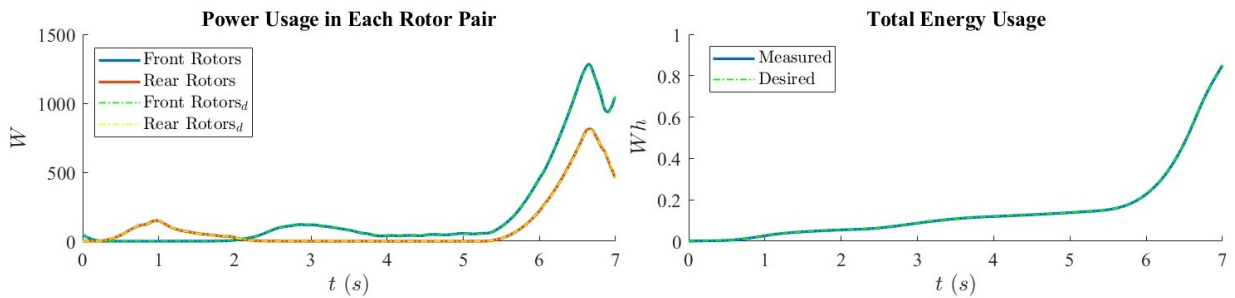


Figure 5.7: Power and energy consumed by the feedback controller compared to the trajectory optimisation solution

Results

To evaluate the performance of the LQR controller, a series of simulations were conducted in which the aircraft's initial horizontal distance, altitude, and/or horizontal velocity were disturbed. The controller's performance was assessed based on its ability to attenuate these disturbances and achieve a stable hover state while minimising the total energy required to transition. To account for rotor inefficiencies, the calculated energy values were scaled by a factor of 1.5 to reflect an approximate rotor efficiency of 65% [7, p. 9]. The results of these experiments are summarised in Table 6.1 with corresponding state, input, and energy consumption graphs presented in Appendix C.

As expected, given the cost matrices (Q and R) defined in the LQR controller, the system consistently prioritised minimising the final pitch angle ($\bar{\theta}$) and pitch rate errors ($\dot{\bar{\theta}}$). This was a key success of the controller, demonstrating its ability to reach a stable hover regardless of the initial disturbance, thus ensuring a reliable handover to the existing control system.

However, the controller exhibited reduced performance when experiencing increased initial velocity. As seen in Sims 5 - 7, the higher initial velocity led to an increase in energy consumption and difficulty during the transition and achieving the desired final position. Conversely, when the initial velocity was reduced, the energy consumption decreased, consistent with the observations from Segment 2 of flight F79 (Table 2.1). Although this model does not include wind effects, the results are encouraging, as all flight logs (F77, F78, and F79) recorded tailwinds where the airspeed experienced by the aircraft is less than the ground speed. Nonetheless, these findings also indicate that the current controller configuration may be unsuitable for landing into a headwind, where the system's safety could be compromised.

Overall, the controller has demonstrated that under tailwind and windless conditions, energy consumption during the landing transition can be reduced by more than 75% by combining the transition to hover with the initial VTOL descent and shortening the total transition time to 7s (assuming an average of 6.9Wh consumed between Waypoint 12 and Point 4 in the current flight landing transitions - Table 2.1). These findings indicate that optimal control represents a highly promising approach for improving the landing trajectory energy efficiency of the Griffin Pro UAV.

Furthermore, this project has developed comprehensive tools for modelling complex nonlinear aerodynamics that produce coupling effects between the aerodynamic control surfaces of hybrid aircraft. These tools have improved the understanding of the Griffin Pro's flight dynamics and may contribute to enhancements in the existing control system, independent of the optimal trajectory developed in this work.

In addition, this research has resulted in a versatile and flexible simulation framework that allows the model, trajectory, and/or controller to be readily modified or redesigned in response to future findings or system developments.

6.1 Future Works

While this initial investigation provided strong support for the implementation of optimal control in the landing transition of the Griffin Pro, further development is required before physical implementation. The project must be extended beyond its current limitations to improve the fidelity of the aircraft model and its representation of real-world dynamics.

6.1.1 Model Verification

The current Griffin Pro aircraft model presented in this thesis incorporates several estimated parameter measurements, including the moment of inertia about the y -axis, the surface areas of the wings and tail, and the distances from the centre of pressure of each aerodynamic control surface to the aircraft's centre of gravity. While these estimations enabled the construction of the model in Chapter 4, they also introduce modelling errors. This was evident in the cruise flight trim analysis, where the resulting equilibrium solution required a non-zero tilt angle of the front rotors. When the Griffin Pro is in cruise flight, the front rotors should be purely horizontal in the body frame, indicating the need for verification and refinement of the physical parameter used in this model. The most effective approach would be to collaborate with ARACE to obtain a technical design specification document for the aircraft. Failing that, parameter estimation could be improved through directly measuring the physical aircraft to verify and refine all parameters. thereby reducing potential sources of modelling error.

6.1.2 Wind Modelling

As discussed in the results, the current model does not incorporate wind effects or account for disturbances arising from wind gust and flow variation outside the feedback controller. Wind can significantly affect the airspeed experienced by aerodynamic control surfaces, leading to errors in the prediction of lift and drag forces during real flight. This issue was observed in flight F79, where the measured airspeed was approximately $5m/s$ less than the groundspeed. Because aerodynamic forces depend on airspeed while aircraft states are referenced to the ground, this discrepancy poses a major concern for any physical testing without wind modelling. Wind disturbance modelling and compensation techniques are well documented [8, 32], and their inclusion would substantially improve the robustness and physical reliability of the Griffin Pro model.

6.1.3 Lateral Dynamics Analysis

The current work was conducted under the assumption that the Griffin Pro's existing lateral controller can stabilise the lateral dynamics during the landing manoeuvre. However, this assumption requires verification. The present model considers only the longitudinal dynamics that inherently assumes uniform airspeed across both winds, the rotors within a pair are symmetrical. in real flight, these assumptions may not hold under many operating conditions. A further investigation is therefore needed to determine whether the inclusion of lateral dynamics is necessary for accurate modelling and safe controller implementation. This analysis would benefit from consultation with experts on the current Griffin Pro control for a deeper understanding of the Ardupilot control architecture used in the Griffin Pro.

6.1.4 Ardupilot Implementation

The Griffin Pro is currently automated via Ardupilot software; therefore, the optimal control developed in this project must ultimately be implemented within Ardupilot before any physical testing can occur. The state and input control was assigned to align with how the aircraft is expected to be represented in Ardupilot hopefully facilitating a simple integration by mapping the states and inputs in this model with the same states and inputs in Ardupilot. Following software integration, the control system should be validated in Realfight's simulation environment to assess its performance and safety in a third party physics model before real-world testing.

6.2 Conclusion

This initial investigation has demonstrated that optimal control offers a promising pathway for improving the energy efficiency of the Griffin Pro's landing trajectory. The project successfully identified inefficiencies in the current system, developed a longitudinal dynamic model of the Griffin Pro capable of predicting complex aerodynamic forces, formulated an optimal trajectory that reduces transition distance, duration, and VTOL requirements while minimising energy consumption, and implemented a controller that was designed to ensure stable convergence and prioritise aircraft safety.

Although further work is required to enhance the model's fidelity and enable physical implementation, this research provides a strong foundational framework for future development. The tools, methodologies, and insights established in this project will support continued progress toward a fully integrated optimal control system for the Griffin Pro, with the potential to significantly improve operational efficiency of the landing phase.

Test	Initial State			Final State				Energy _T
Sim 0	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−50	20	20	−0.01	0.02	0.01	1.88	0.851
Disturbance %	0	0	0				In Hover	−1.53
Sim 1	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−55	22	19	−5.64	0.55	0.07	1.88	0.833
Disturbance %	10	10	−5	112.8	27.7	7.1	In Hover	−3.61
Sim 2	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−45	18	18	−6.18	0.54	0.23	1.91	0.840
Disturbance %	−10	−10	−10	123.6	27.1	11.7	In Hover	−2.74
Sim 3	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−60	18	18.5	−8.11	0.39	0.36	1.90	0.818
Disturbance %	20	−10	−7.5	81.1	19.4	24.2	In Hover	−5.35
Sim 4	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−45	25	18.5	2.91	1.33	0.05	1.88	0.846
Disturbance %	−10	25	−7.5	−58.3	26.7	3.1	In Hover	−2.05
Sim 5	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−50	20	21	2.00	1.71	0.06	1.88	0.894
Disturbance %	0	0	5	overshoot		5.8	In Hover	3.51
Sim 6	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−55	25	21	15.03	19.29	1.11	1.67	1.29
Disturbance %	10	25	5	−300.6	385.9	110.9	No Hover	49.01
Sim 7	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−55	25	20.5	3.57	9.74	0.33	1.86	0.972
Disturbance %	10	25	2.5	−71.5	194.8	65.4	In Hover	12.54
Sim 8	x (m)	h (m)	$\dot{x}$ (m/s)	x (m)	h (m)	V (m/s)	$ \theta + \dot{\theta} $ (°)	Wh
Value	−55	25	15	−5.65	3.07	0.43	1.24	0.846
Disturbance %	10	25	−25	113.0	61.4	8.6	In Hover	−2.05

Table 6.1: Simulation data observing the ability of the controller to safely converge the system when the initial position is disturbed

Bibliography

- [1] ChatGPT, . URL <https://chatgpt.com/>.
- [2] CLARK Y AIRFOIL (clarky-il), . URL <http://airfoiltools.com/airfoil/details?airfoil=clarky-il>.
- [3] Fliight Technologies, . URL <https://fliight.com.au>.
- [4] Drone Types: Multi-Rotor, Fixed-Wing, Single Rotor, Hybrid VTOL, Nov. 2025. URL <https://www.auav.com.au/articles/drone-types/>. Section: Articles.
- [5] G. Allibert, D. Abeywardena, M. Bangura, and R. Mahony. Estimating body-fixed frame velocity and attitude from inertial measurements for a quadrotor vehicle. In *2014 IEEE Conference on Control Applications (CCA)*, pages 978–983, Juan Les Antibes, France, Oct. 2014. IEEE. ISBN 978-1-4799-7409-2. doi: 10.1109/CCA.2014.6981462. URL <https://ieeexplore.ieee.org/document/6981462>.
- [6] ARACE. Griffin Pro Spec Sheet. URL <https://thegooddroneco.co.za/wp-content/uploads/2024/03/231207-ARACE-Griffin-Pro.pdf>.
- [7] M. Bangura, M. Melega, R. Naldi, and R. Mahony. Aerodynamics of Rotor Blades for Quadrotors, Jan. 2016. URL <http://arxiv.org/abs/1601.00733>. arXiv:1601.00733 [physics].
- [8] R. W. Beard and T. W. McLain. *Theory and Practice*. Princeton University Press, Princeton, 2012. ISBN 978-1-4008-4060-1. doi: doi:10.1515/9781400840601. URL <https://doi.org/10.1515/9781400840601>.
- [9] J. T. Betts. *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming, Second Edition*. Society for Industrial and Applied Mathematics, second edition, 2010. doi: 10.1137/1.9780898718577. URL <https://epubs.siam.org/doi/abs/10.1137/1.9780898718577>. eprint: <https://epubs.siam.org/doi/pdf/10.1137/1.9780898718577>.
- [10] R. Bordalba, T. Schoels, L. Ros, J. M. Porta, and M. Diehl. Direct Collocation Methods for Trajectory Optimization in Constrained Robotic Systems. *IEEE Transactions on Robotics*, 39 (1):183–202, Feb. 2023. ISSN 1552-3098, 1941-0468. doi: 10.1109/TRO.2022.3193776. URL <http://arxiv.org/abs/2304.12908>. arXiv:2304.12908 [cs].
- [11] R. Cory and R. Tedrake. Experiments in Fixed-Wing UAV Perching. In *AIAA Guidance, Navigation and Control Conference and Exhibit*, Honolulu, Hawaii, Aug. 2008. American Institute of Aeronautics and Astronautics. ISBN 978-1-60086-999-0. doi: 10.2514/6.2008-7256. URL <https://arc.aiaa.org/doi/10.2514/6.2008-7256>.
- [12] F. Giraud and C. Giraud-Audine. Chapter Four - Control in the rotating reference frame. In F. Giraud and C. Giraud-Audine, editors, *Piezoelectric Actuators: Vector Control Method*, pages 137–160. Butterworth-Heinemann, 2019. ISBN 978-0-12-814186-1. doi: <https://doi.org/10.1016/>

- B978-0-12-814186-1.00008-9. URL <https://www.sciencedirect.com/science/article/pii/B9780128141861000089>.
- [13] S. Gudmundsson. Chapter 14 - The Anatomy of the Propeller. In S. Gudmundsson, editor, *General Aviation Aircraft Design*, pages 581–659. Butterworth-Heinemann, Boston, 2014. ISBN 978-0-12-397308-5. doi: <https://doi.org/10.1016/B978-0-12-397308-5.00014-3>. URL <https://www.sciencedirect.com/science/article/pii/B9780123973085000143>.
- [14] J. k. Gunarathna and R. Munasinghe. Development of a Quad-rotor Fixed-wing Hybrid Unmanned Aerial Vehicle. In *2018 Moratuwa Engineering Research Conference (MERCon)*, pages 72–77, May 2018. doi: [10.1109/MERCon.2018.8421941](https://doi.org/10.1109/MERCon.2018.8421941). URL <https://ieeexplore.ieee.org/document/8421941/>.
- [15] H. He and H. Duan. A multi-strategy pigeon-inspired optimization approach to active disturbance rejection control parameters tuning for vertical take-off and landing fixed-wing UAV. *Chinese Journal of Aeronautics*, 35(1):19–30, Jan. 2022. ISSN 10009361. doi: [10.1016/j.cja.2021.05.010](https://doi.org/10.1016/j.cja.2021.05.010). URL <https://linkinghub.elsevier.com/retrieve/pii/S1000936121001898>.
- [16] I. Heiets, Y.-W. Kuo, J. La, R. C. K. Yeun, and W. Verhagen. Future Trends in UAV Applications in the Australian Market. *Aerospace*, 10(6):555, June 2023. ISSN 2226-4310. doi: [10.3390/aerospace10060555](https://doi.org/10.3390/aerospace10060555). URL <https://www.mdpi.com/2226-4310/10/6/555>. Publisher: Multidisciplinary Digital Publishing Institute.
- [17] J. Hu, J. Wei, K. Liu, X. Yu, M. Cao, and Z. Qin. Hybrid Mode: Routinization of the Transition Mode as the Third Common Mode for Compound VTOL Drones. *Drones*, 8(3):93, Mar. 2024. ISSN 2504-446X. doi: [10.3390/drones8030093](https://doi.org/10.3390/drones8030093). URL <https://www.mdpi.com/2504-446X/8/3/93>. Publisher: Multidisciplinary Digital Publishing Institute.
- [18] M. Kelly. An Introduction to Trajectory Optimization: How to Do Your Own Direct Collocation. *SIAM Review*, 59(4):849–904, Jan. 2017. ISSN 0036-1445, 1095-7200. doi: [10.1137/16M1062569](https://doi.org/10.1137/16M1062569). URL <https://epubs.siam.org/doi/10.1137/16M1062569>.
- [19] M. Kelly, C. McGreal, and Cmcgreal-Swift. MatthewPeterKelly/OptimTraj: First stable release, Dec. 2022. URL <https://zenodo.org/record/7430524>.
- [20] M. L. Latash and V. M. Zatsiorsky. 10 - Redundancy and Abundance. In M. L. Latash and V. M. Zatsiorsky, editors, *Biomechanics and Motor Control*, pages 177–204. Academic Press, San Diego, 2016. ISBN 978-0-12-800384-8. doi: <https://doi.org/10.1016/B978-0-12-800384-8.00010-7>. URL <https://www.sciencedirect.com/science/article/pii/B9780128003848000107>.
- [21] G. J. Leishman. *Principles of Helicopter Aerodynamics with CD Extra*. Cambridge University Press, Apr. 2006. ISBN 978-0-521-85860-1.
- [22] MatthewPeterKelly. OptimTraj/docs at master · MatthewPeterKelly/OptimTraj, Mar. 2018. URL <https://github.com/MatthewPeterKelly/OptimTraj/tree/master/docs>.
- [23] K. W. McAlister, C. A. Schuler, L. Branum, and J. C. Wu. 3-D Wake Measurements Near a Hovering Rotor for Determining Profile and Induced Drag. Aug. 1995.

- [24] A. Mitra, S. Sarkar, and A. Chakraborty. 15 - Multibody dynamic analysis of onshore horizontal-axis wind turbine. In F. P. G. Marquez, M. Papaelias, and V. L. J. Junior, editors, *Non-Destructive Testing and Condition Monitoring Techniques in Wind Energy*, Wind Energy Engineering, pages 351–397. Academic Press, 2023. ISBN 978-0-323-99666-2. doi: <https://doi.org/10.1016/B978-0-323-99666-2.00015-0>. URL <https://www.sciencedirect.com/science/article/pii/B9780323996662000150>.
- [25] S. Omari, M.-D. Hua, G. Ducard, and T. Hamel. Nonlinear control of VTOL UAVs incorporating flapping dynamics. In *2013 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 2419–2425, Nov. 2013. doi: 10.1109/IROS.2013.6696696. URL <https://ieeexplore.ieee.org/document/6696696/>. ISSN: 2153-0866.
- [26] O. Procházka, F. Novák, T. Báča, P. M. Gupta, R. Pěnička, and M. Saska. Model predictive control-based trajectory generation for agile landing of unmanned aerial vehicle on a moving boat. *Ocean Engineering*, 313:119164, 2024. ISSN 0029-8018. doi: <https://doi.org/10.1016/j.oceaneng.2024.119164>. URL <https://www.sciencedirect.com/science/article/pii/S0029801824025022>.
- [27] J. W. Roberts, R. Cory, and R. Tedrake. On the controllability of fixed-wing perching. In *2009 American Control Conference*, pages 2018–2023, St. Louis, MO, USA, 2009. IEEE. ISBN 978-1-4244-4523-3. doi: 10.1109/ACC.2009.5160526. URL <http://ieeexplore.ieee.org/document/5160526/>.
- [28] A. S. Saeed, A. B. Younes, C. Cai, and G. Cai. A survey of hybrid Unmanned Aerial Vehicles. *Progress in Aerospace Sciences*, 98:91–105, Apr. 2018. ISSN 03760421. doi: 10.1016/j.paerosci.2018.03.007. URL <https://linkinghub.elsevier.com/retrieve/pii/S0376042117302233>.
- [29] R. F. Stengel. *Flight Dynamics*. Princeton University Press, Princeton, 2005. ISBN 978-1-4008-6681-6. doi: doi:10.1515/9781400866816. URL <https://doi.org/10.1515/9781400866816>.
- [30] R. Tedrake. *Underactuated Robotics: Algorithms for Walking, Running, Swimming, Flying, and Manipulation*. Course Notes for MIT 6.832, 2023. URL <https://underactuated.csail.mit.edu>.
- [31] A. Wickenheiser, E. Garcia, and M. Waszak. Longitudinal dynamics of a perching aircraft concept. page 192, San Diego, CA, May 2005. doi: 10.1117/12.601512. URL <http://proceedings.spiedigitallibrary.org/proceeding.aspx?doi=10.1117/12.601512>.
- [32] J. Willis, J. Johnson, and R. W. Beard. State-Dependent LQR Control for a Tilt-Rotor UAV. In *2020 American Control Conference (ACC)*, pages 4175–4181, July 2020. doi: 10.23919/ACC45564.2020.9147931. URL <https://ieeexplore.ieee.org/document/9147931/>. ISSN: 2378-5861.
- [33] J. B. Willis and R. W. Beard. Pitch and Thrust Allocation for Full-Flight-Regime Control of Winged eVTOL UAVs. *IEEE Control Systems Letters*, 6:1058–1063, 2022. ISSN 2475-1456. doi: 10.1109/LCSYS.2021.3089130. URL <https://ieeexplore.ieee.org/document/9454502/>.

Appendix A: Griffin Pro Flight Data

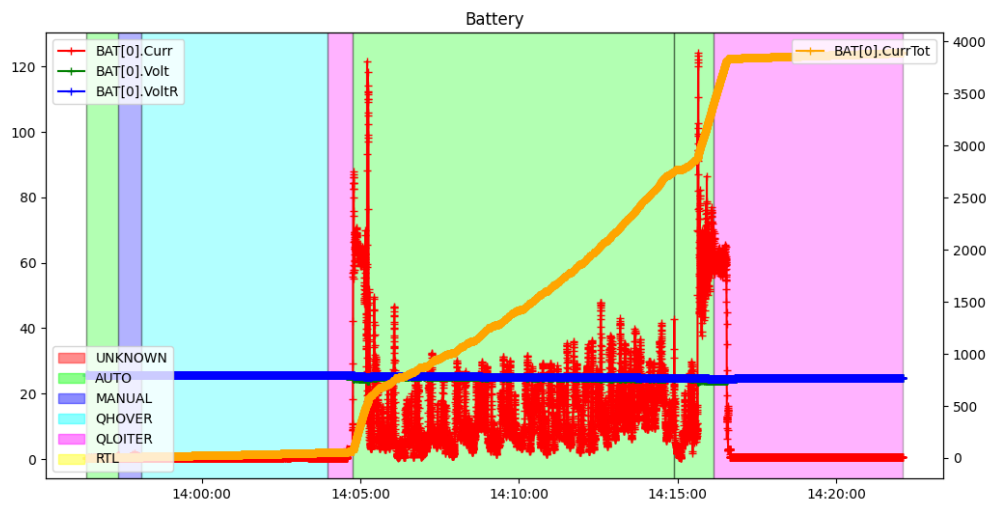


Figure A.1: Battery measurement during flight F77

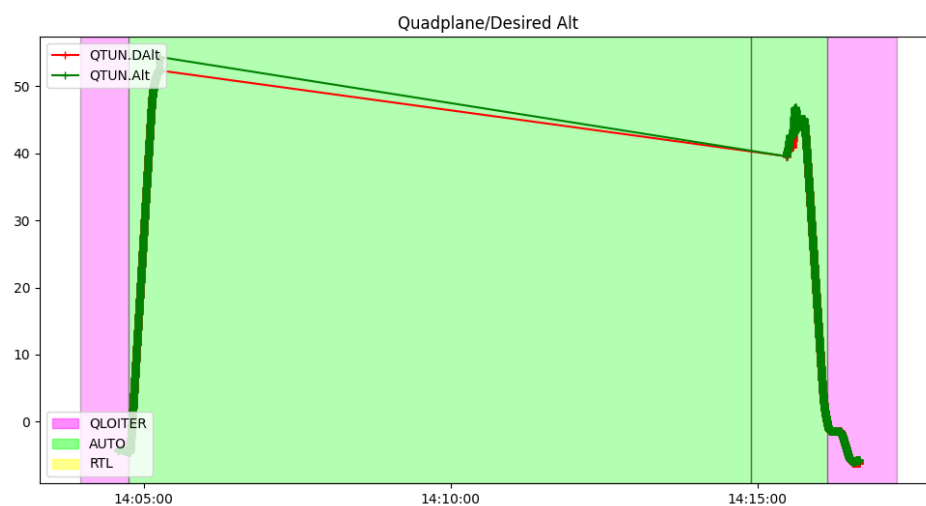


Figure A.2: Desired altitude measurement during flight F77

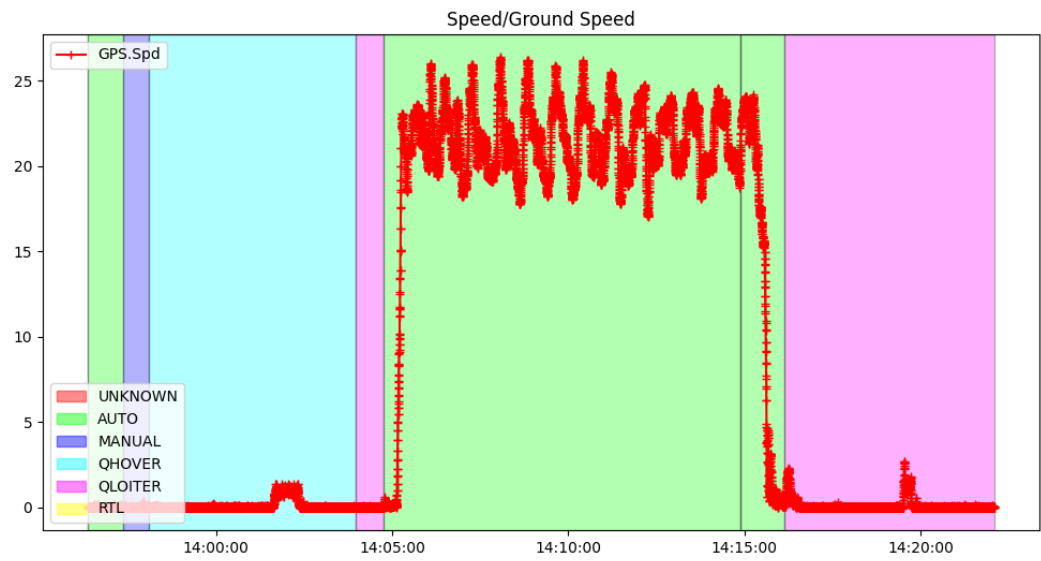


Figure A.3: Groundspeed measurement during flight F77

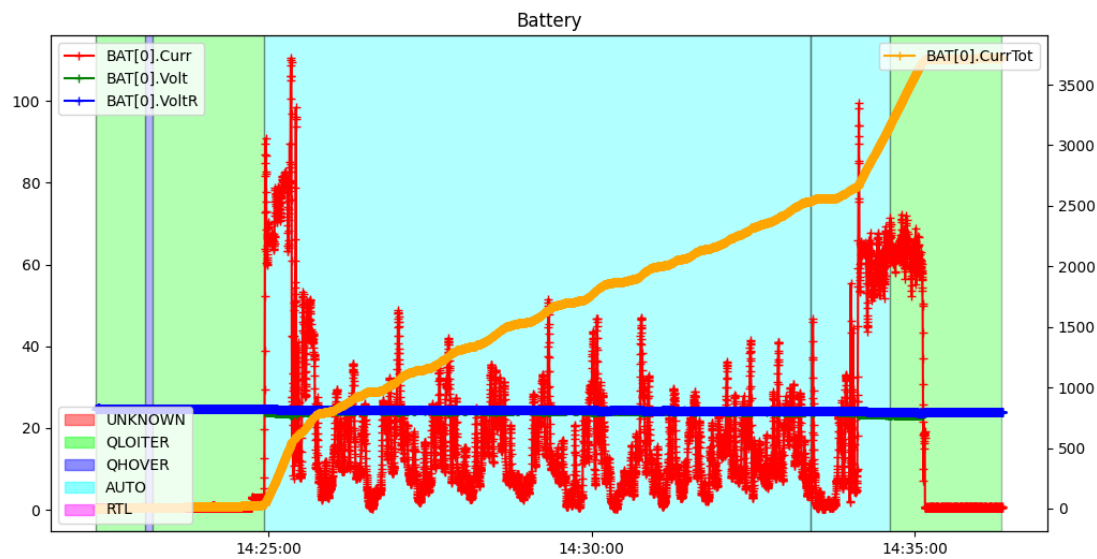


Figure A.4: Battery measurement during flight F78

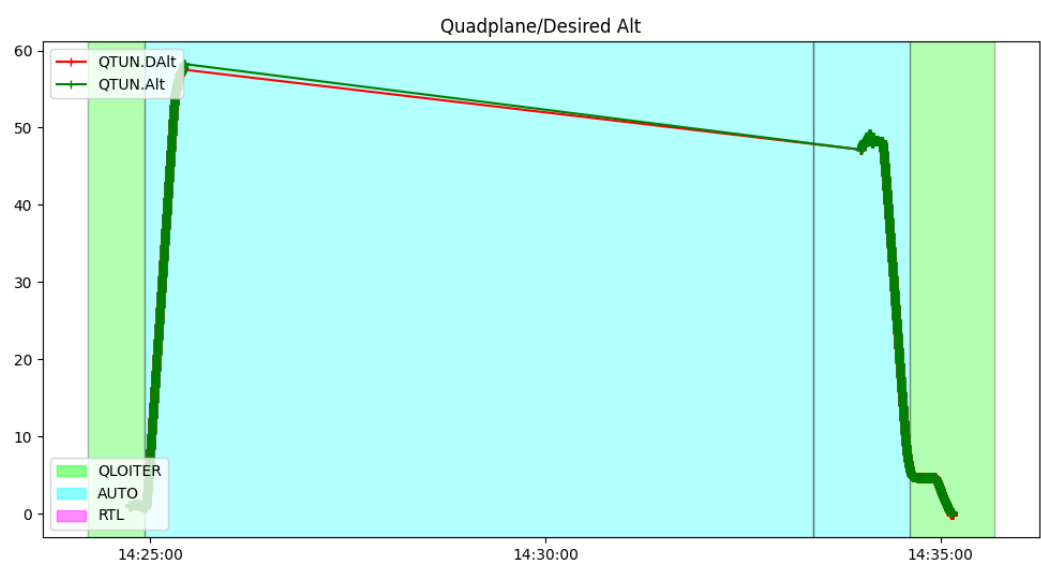


Figure A.5: Desired altitude measurement during flight F78

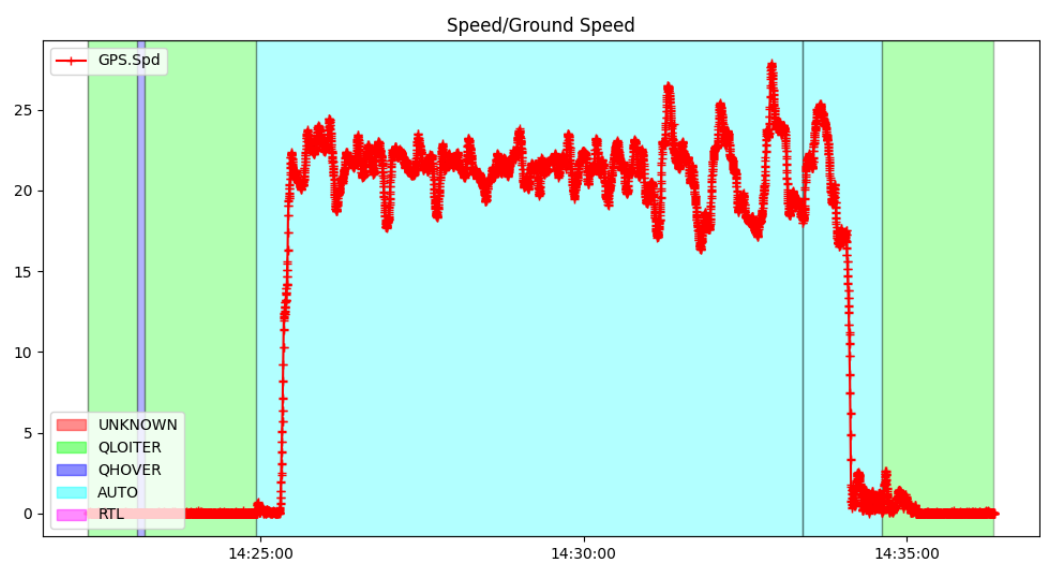


Figure A.6: Groundspeed measurement during flight F78

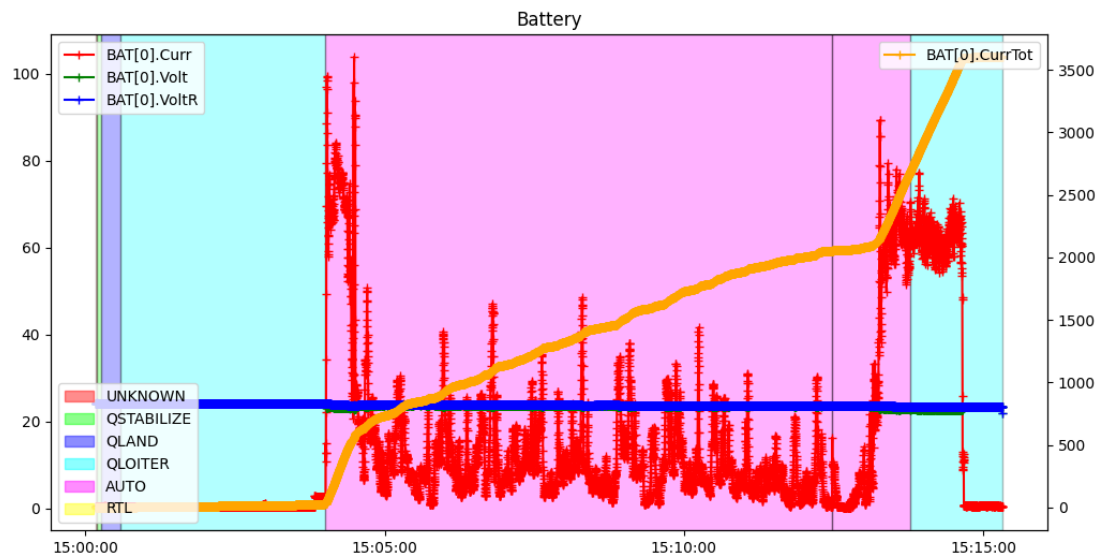


Figure A.7: Battery measurement during flight F79

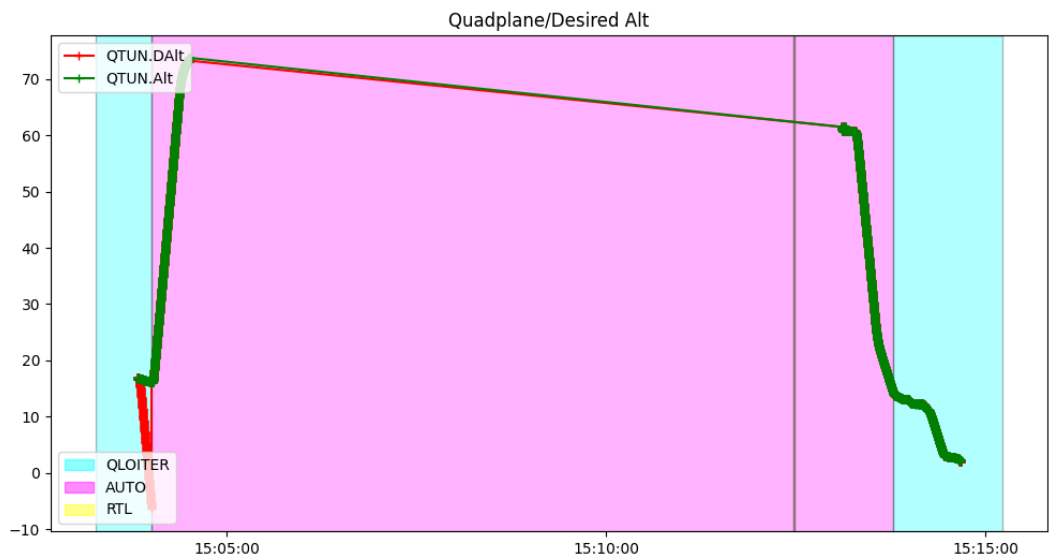


Figure A.8: Desired altitude measurement during flight F79

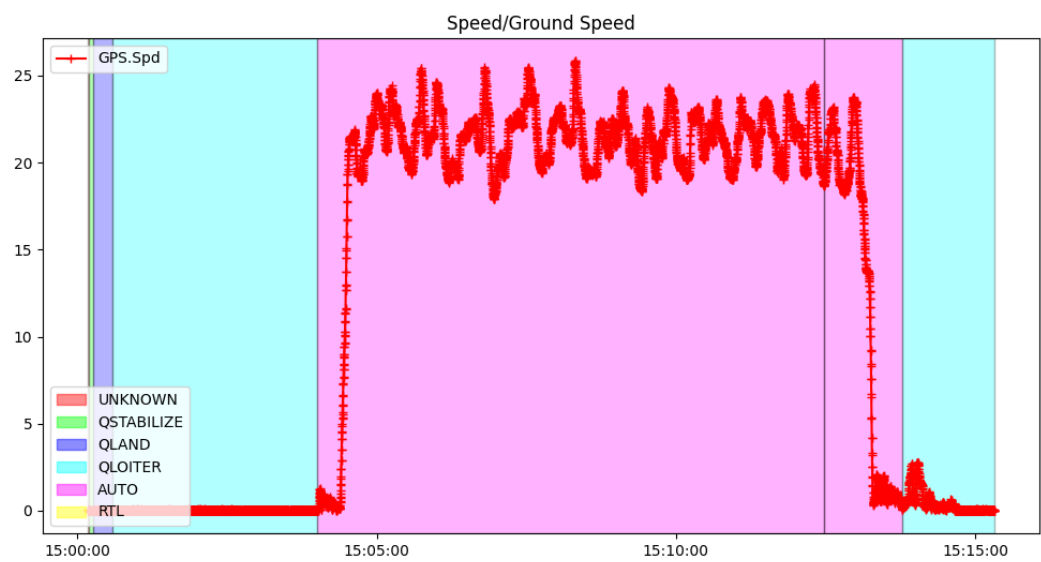


Figure A.9: Groundspeed measurement during flight F79

Appendix B: Rotor Power Derivation

These equations were expanded and simplified with assistance from ChatGPT[1]. From Equations 4.17, 4.18, and 4.19, we have:

$$L_r = 2\rho\pi R_r^2 v_z^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (\text{B.1})$$

$$D_r = 2\rho\pi R_r^2 v_x^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (\text{B.2})$$

$$P_r = L_r(v_z^i - V_{vr}) + D_r(v_x^i + V_{hr}). \quad (\text{B.3})$$

By adding rotor lift and drag forces developed in Equations 3.4, the three equations become:

$$\frac{1}{4}N_b\rho c R_r^3 \bar{\omega}_r^2 C_T = 2\rho\pi R_r^2 v_z^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (\text{B.4})$$

$$\frac{1}{4}N_b\rho c R_r^3 \bar{\omega}_r^2 C_T \mu_r = 2\rho\pi R_r^2 v_x^i \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}, \quad (\text{B.5})$$

$$P_r = \frac{1}{4}N_b\rho c R_r^3 \bar{\omega}_r^2 C_T (v_z^i - V_{vr}) + \frac{1}{4}N_b\rho c R_r^3 \bar{\omega}_r^2 C_T \mu_r (v_x^i + V_{hr}). \quad (\text{B.6})$$

letting $V_a = \sqrt{(v_x^i + V_{hr})^2 + (v_z^i - V_{vr})^2}$, v_z^i and v_x^i can be isolated on the LHS:

$$v_z^i = \frac{N_b c R_r C_T \bar{\omega}_r^2}{8\pi V_a}, \quad (\text{B.7})$$

$$\begin{aligned} v_x^i &= \frac{N_b c R_r C_T \bar{\omega}_r^2}{8\pi V_a} \mu_r \\ &= \mu_r v_z^i, \end{aligned} \quad (\text{B.8})$$

$$P_r = \frac{1}{4}N_b\rho c R_r^3 C_T \bar{\omega}_r^2 (v_z^i - V_{vr} + \mu_r(v_x^i + V_{hr})). \quad (\text{B.9})$$

Beginning with Equation B.7, removing the V_a substitution, implementing the relationship derived in Equation B.8, and letting the vehicle parameters be represented by $K = \frac{N_b c R_r C_T}{8\pi}$, one has:

$$v_z^i = \frac{K \bar{\omega}_r^2}{\sqrt{(\mu_r v_z^i + V_{hr})^2 + (v_z^i - V_{vr})^2}}, \quad (\text{B.10})$$

Expanding and simplifying in terms of v_z^i :

$$\begin{aligned} v_z^{i2} &= \frac{K^2 \bar{\omega}_r^4}{(\mu_r v_z^i + V_{hr})^2 + (v_z^i - V_{vr})^2} \\ v_z^{i2} (\mu_r v_z^i + V_{hr})^2 + (v_z^i - V_{vr})^2 &= K^2 \bar{\omega}_r^4 \\ v_z^{i2} (\mu_r^2 v_z^{i2} + 2\mu_r V_{hr} v_z^i + V_{hr}^2 + v_z^{i2} - 2V_{vr} v_z^i + V_{vr}^2) &= K^2 \bar{\omega}_r^4 \\ \mu_r^2 v_z^{i4} + 2\mu_r V_{hr} v_z^{i3} + V_{hr}^2 v_z^{i2} + v_z^{i4} - 2V_{vr} v_z^{i3} + V_{vr}^2 v_z^{i2} &= K^2 \bar{\omega}_r^4 \\ (1 + \mu_r^2) v_z^{i4} + (2\mu_r V_{hr} - 2V_{vr}) v_z^{i3} + (V_{hr}^2 + V_{vr}^2) v_z^{i2} &= K^2 \bar{\omega}_r^4, \end{aligned} \quad (\text{B.11})$$

the quartic equation with root solution for v_z^i can be expressed:

$$(1 + \mu_r^2)v_z^{i4} + 2(\mu_r V_{h_r} - V_{v_r})v_z^{i3} + (V_{h_r}^2 + V_{v_r}^2)v_z^{i2} - K^2\bar{\omega}_r^4 = 0. \quad (\text{B.12})$$

Using Equation B.8, with the inclusion of $\Lambda = \frac{1}{4}N_b\rho cR_r^3C_T$, Equation B.9 can be refined:

$$\begin{aligned} P_r &= \Lambda\bar{\omega}_r^2(v_z^i - V_{v_r} + \mu_r(\mu_r v_z^i + V_{h_r})) \\ &= \Lambda\bar{\omega}_r^2(v_z^i - V_{v_r} + \mu_r^2 v_z^i + \mu_r V_{h_r}) \\ P_r &= \Lambda\bar{\omega}_r^2((1 + \mu_r^2)v_z^i + \mu_r V_{h_r} - V_{v_r}). \end{aligned} \quad (\text{B.13})$$

To restate the solvable system:

$$(1 + \mu_r^2)v_z^{i4} + 2(\mu_r V_{h_r} - V_{v_r})v_z^{i3} + (V_{h_r}^2 + V_{v_r}^2)v_z^{i2} - K^2\bar{\omega}_r^4 = 0. \quad (\text{B.14})$$

$$v_x^i = \mu_r v_z^i, \quad (\text{B.15})$$

$$P_r = \Lambda\bar{\omega}_r^2((1 + \mu_r^2)v_z^i + \mu_r V_{h_r} - V_{v_r}). \quad (\text{B.16})$$

Appendix C: Disturbance Simulation Graphs

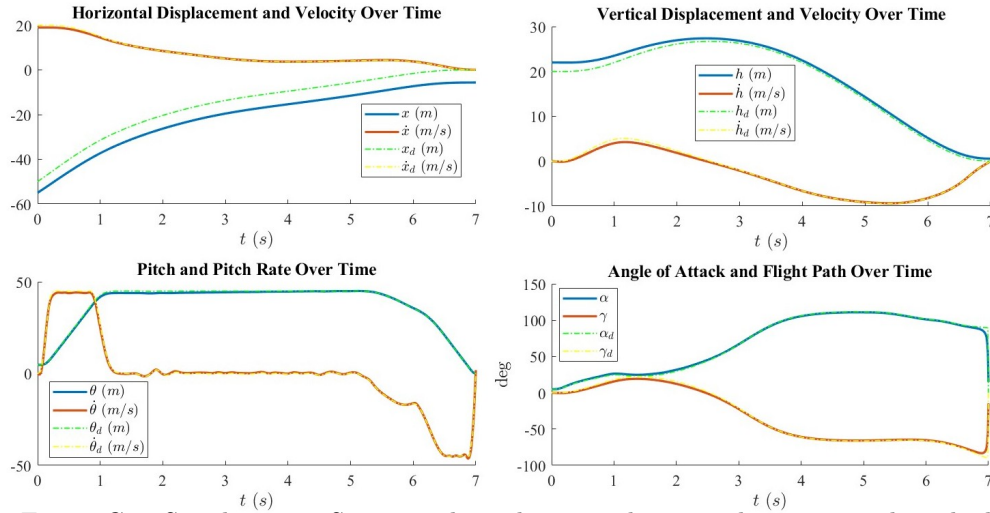


Figure C.1: Simulation 0: States vs desired states when initial position is disturbed.

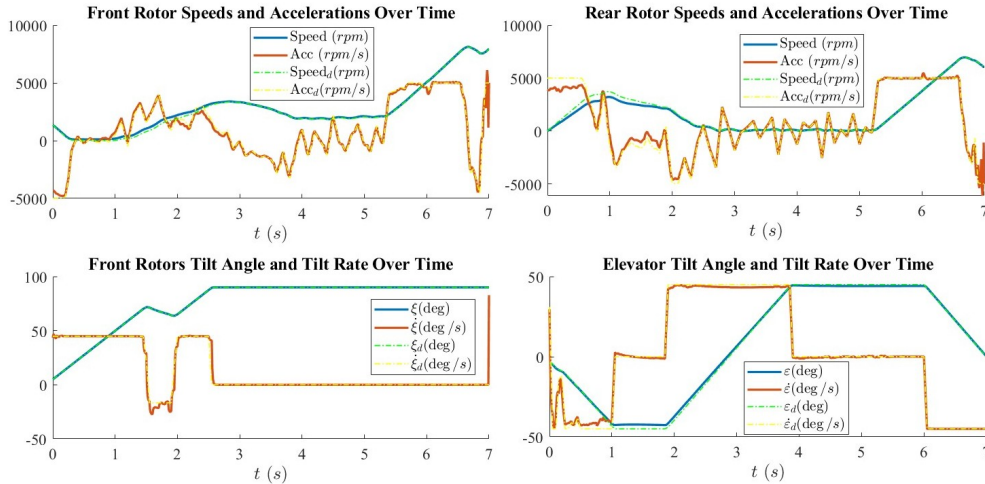


Figure C.2: Simulation 0: Inputs vs desired inputs when initial position is disturbed.

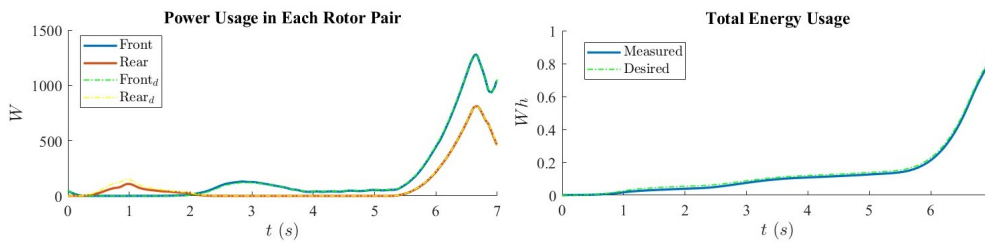


Figure C.3: Simulation 0: Comparison of additional energy consumption.

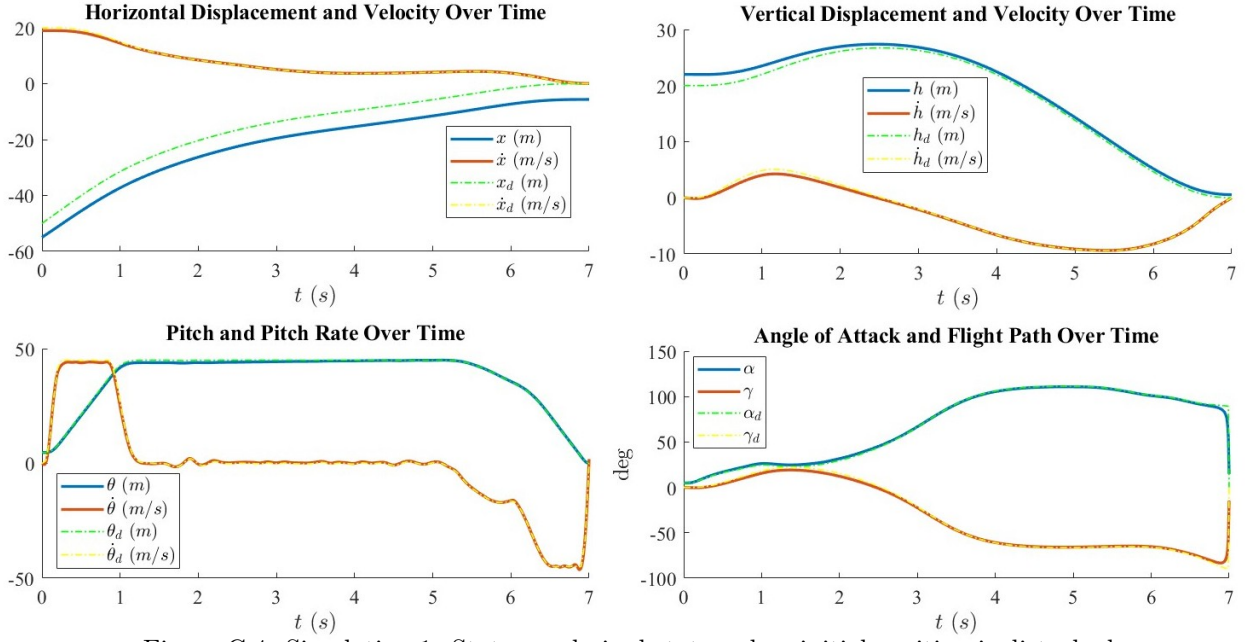


Figure C.4: Simulation 1: States vs desired states when initial position is disturbed.

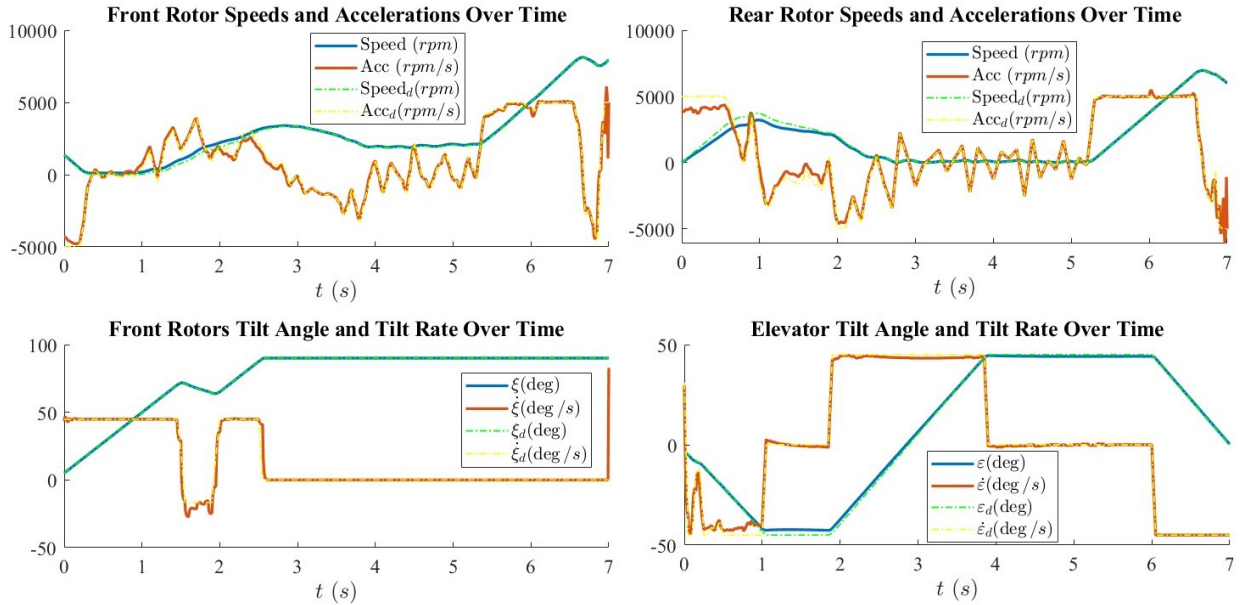


Figure C.5: Simulation 1: Inputs vs desired inputs when initial position is disturbed.

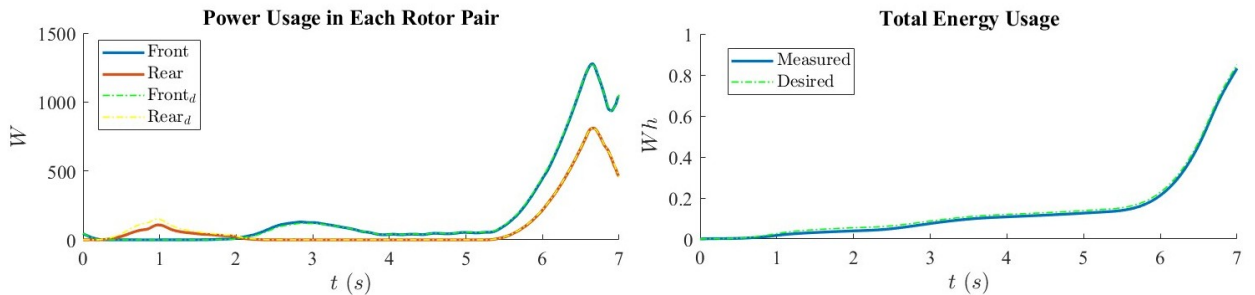


Figure C.6: Simulation 1: Comparison of additional energy consumption.

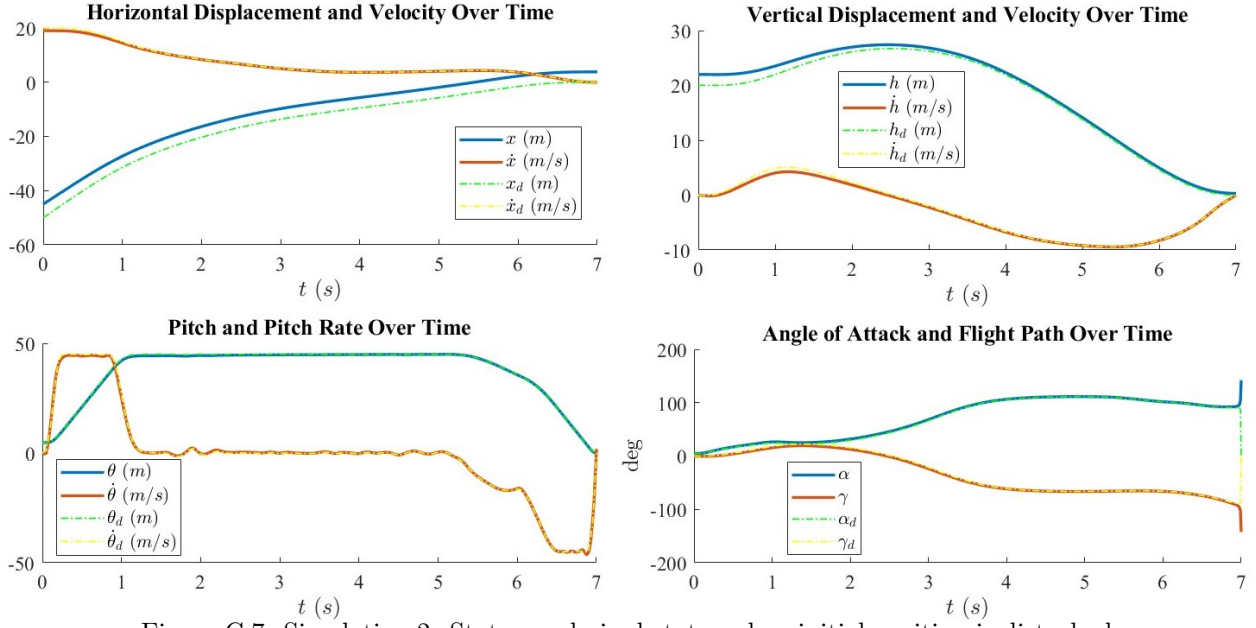


Figure C.7: Simulation 2: States vs desired states when initial position is disturbed.

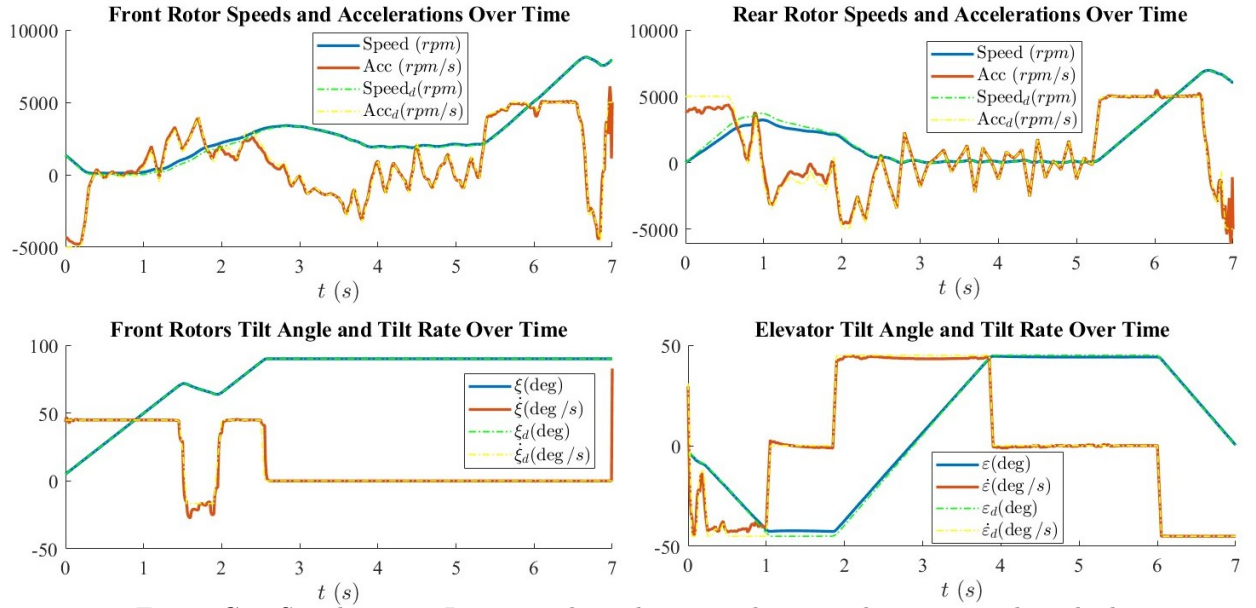


Figure C.8: Simulation 2: Inputs vs desired inputs when initial position is disturbed.

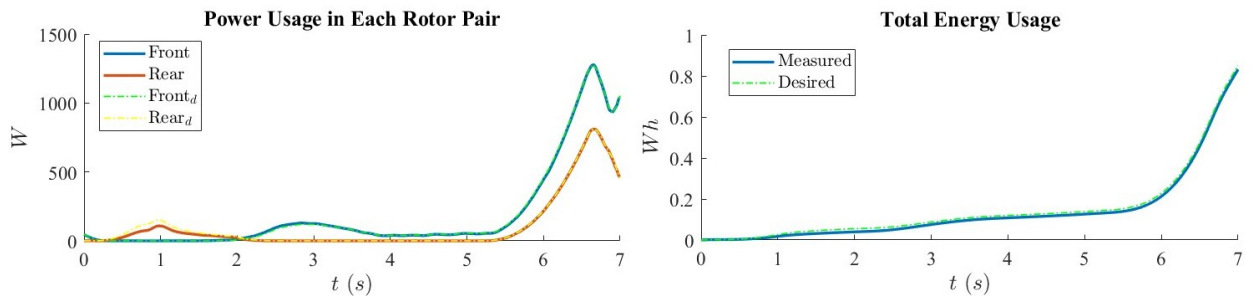


Figure C.9: Simulation 2: Comparison of additional energy consumption.

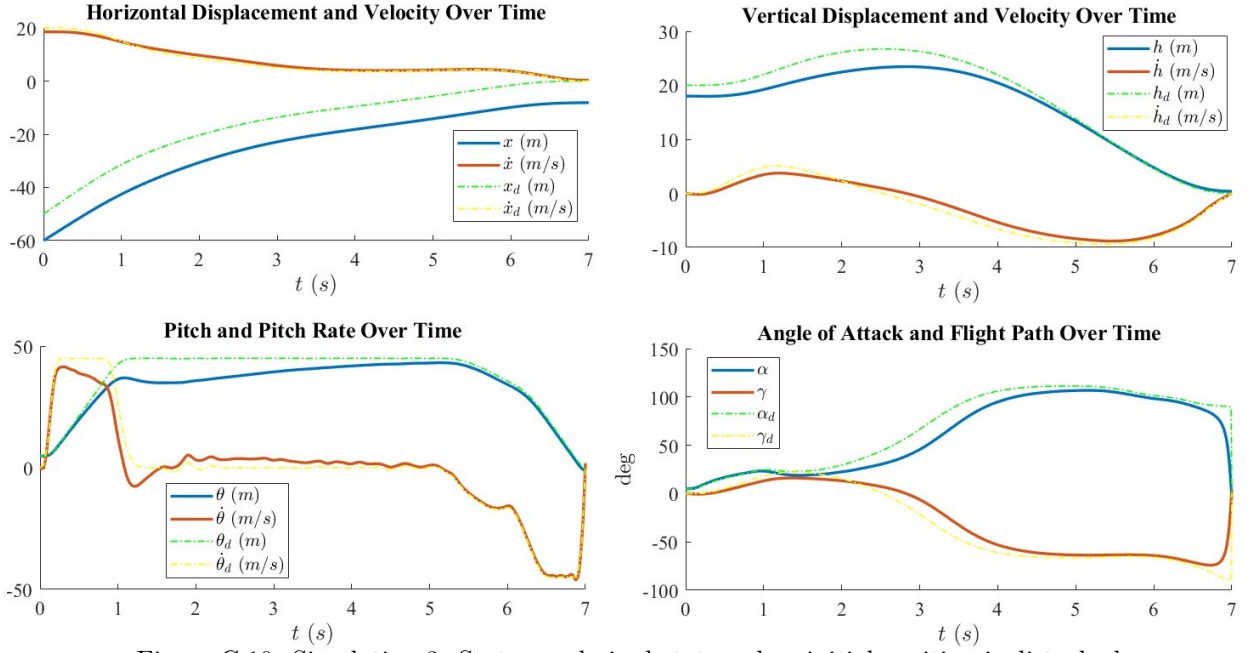


Figure C.10: Simulation 3: States vs desired states when initial position is disturbed.

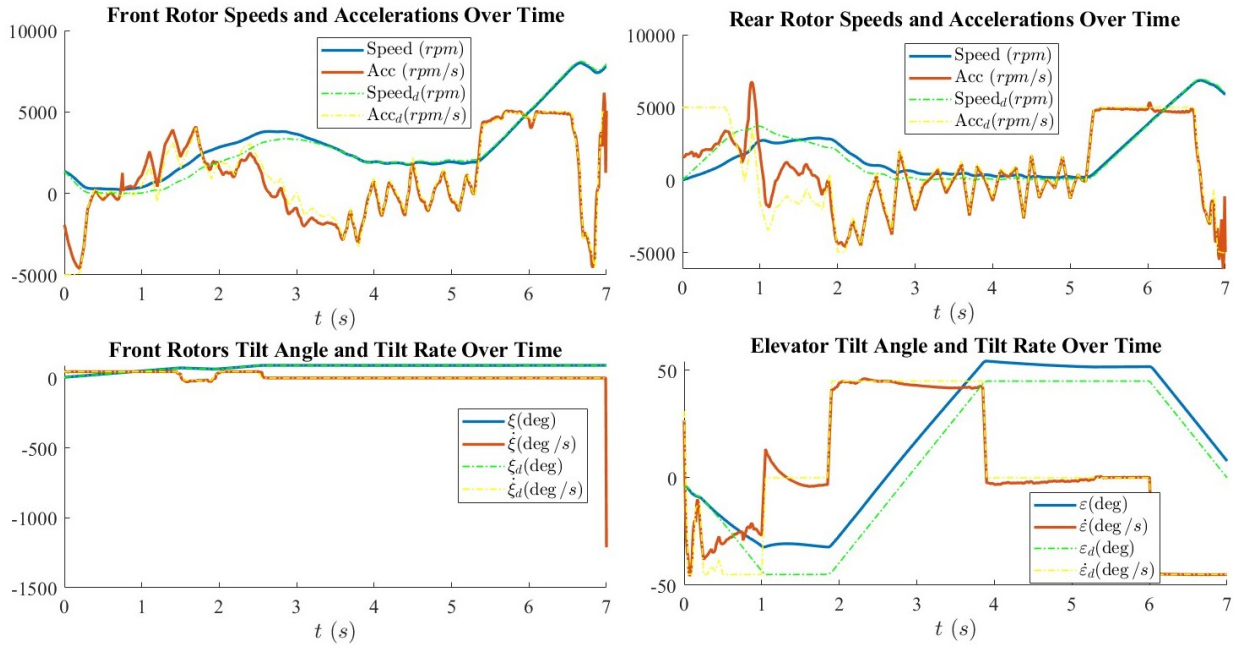


Figure C.11: Simulation 3: Inputs vs desired inputs when initial position is disturbed.

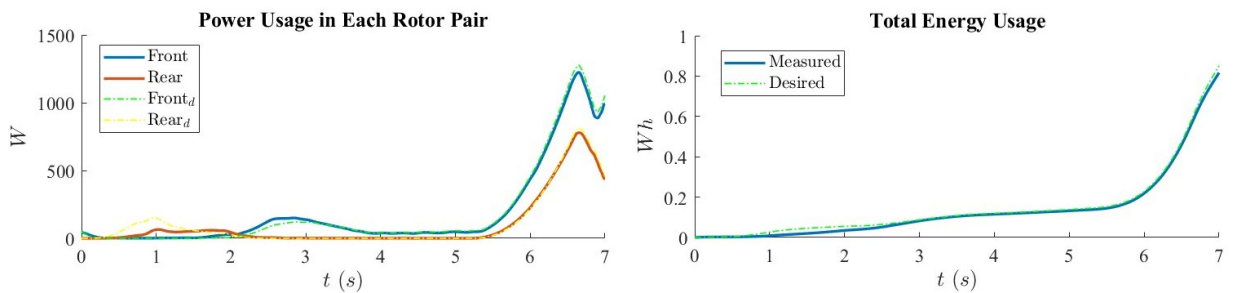
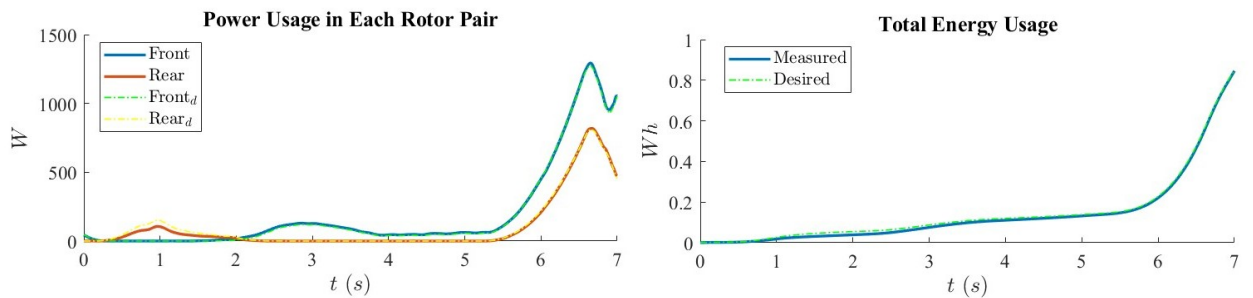
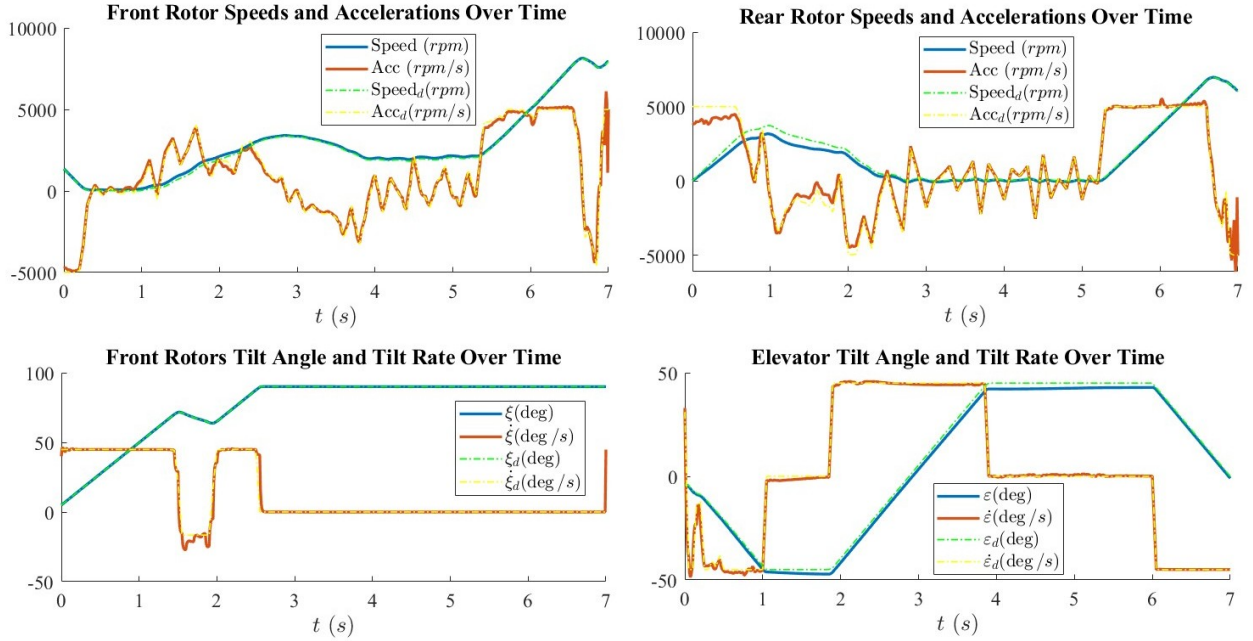
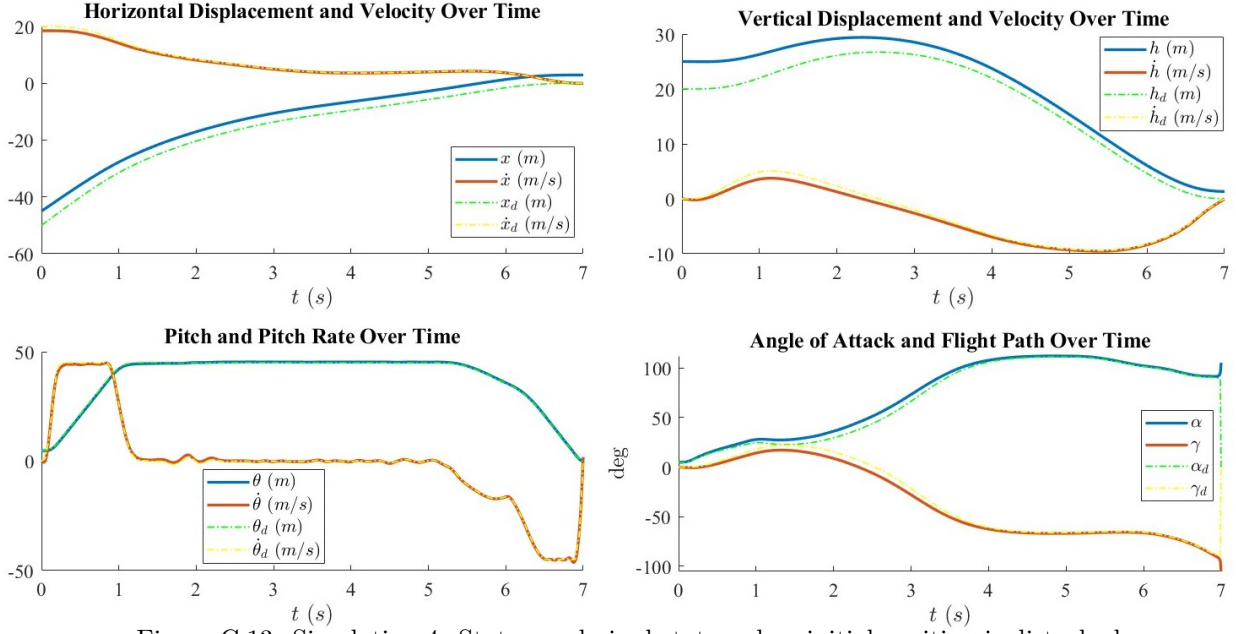


Figure C.12: Simulation 3: Comparison of additional energy consumption.



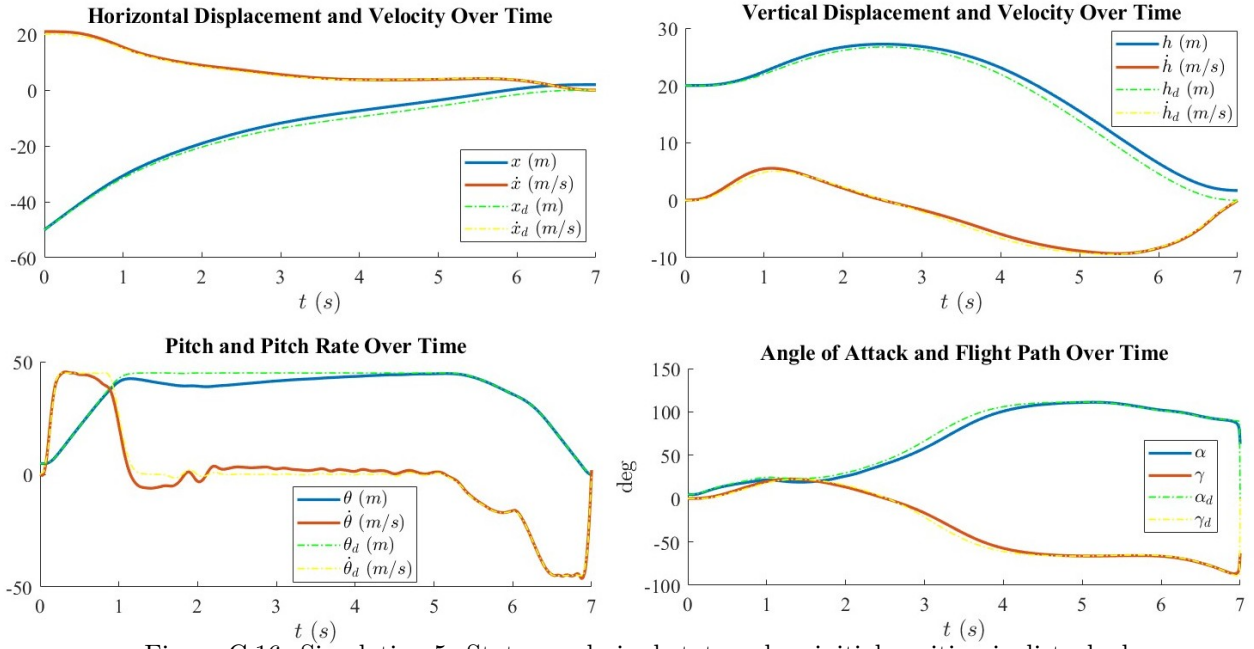


Figure C.16: Simulation 5: States vs desired states when initial position is disturbed.

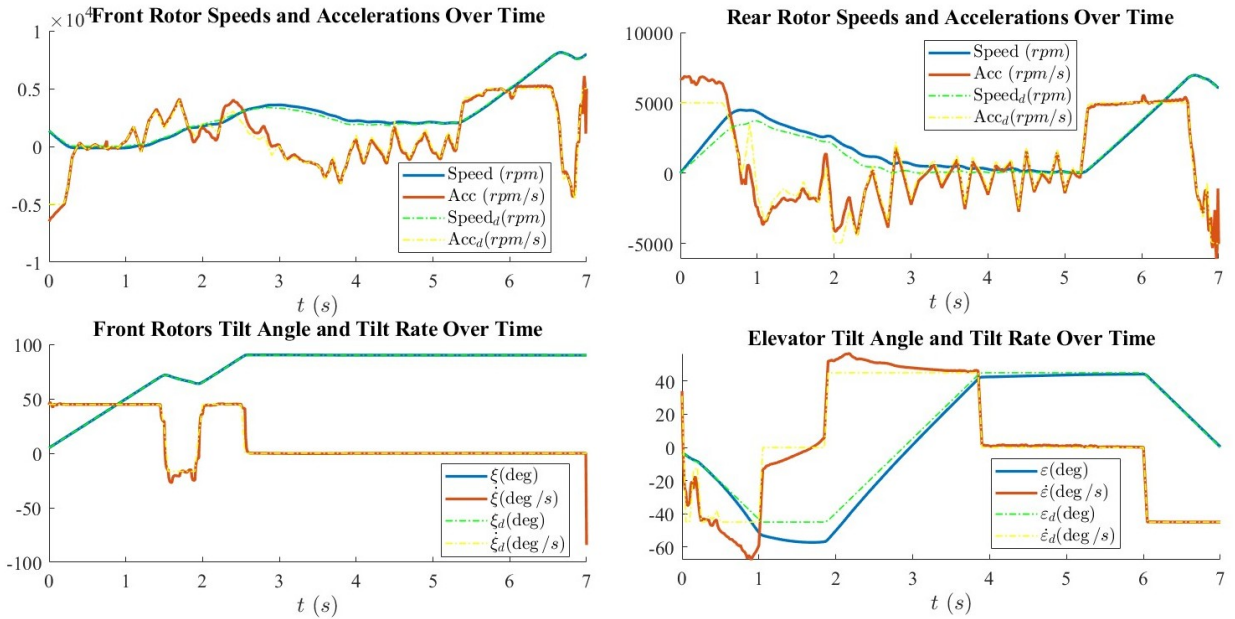


Figure C.17: Simulation 5: Inputs vs desired inputs when initial position is disturbed.

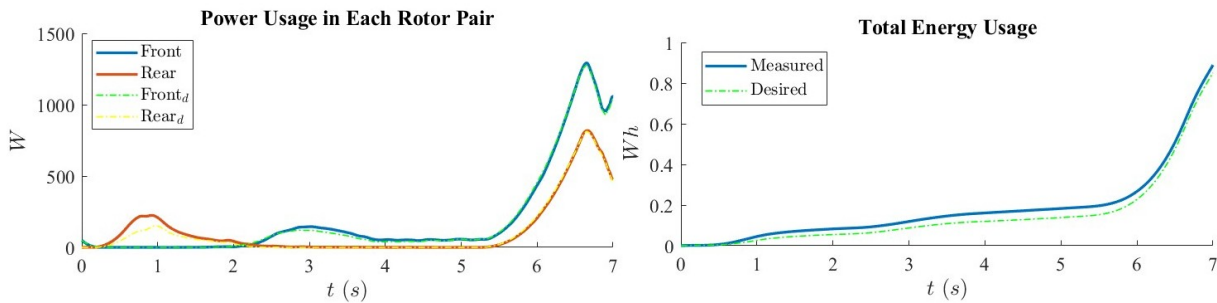


Figure C.18: Simulation 5: Comparison of additional energy consumption.

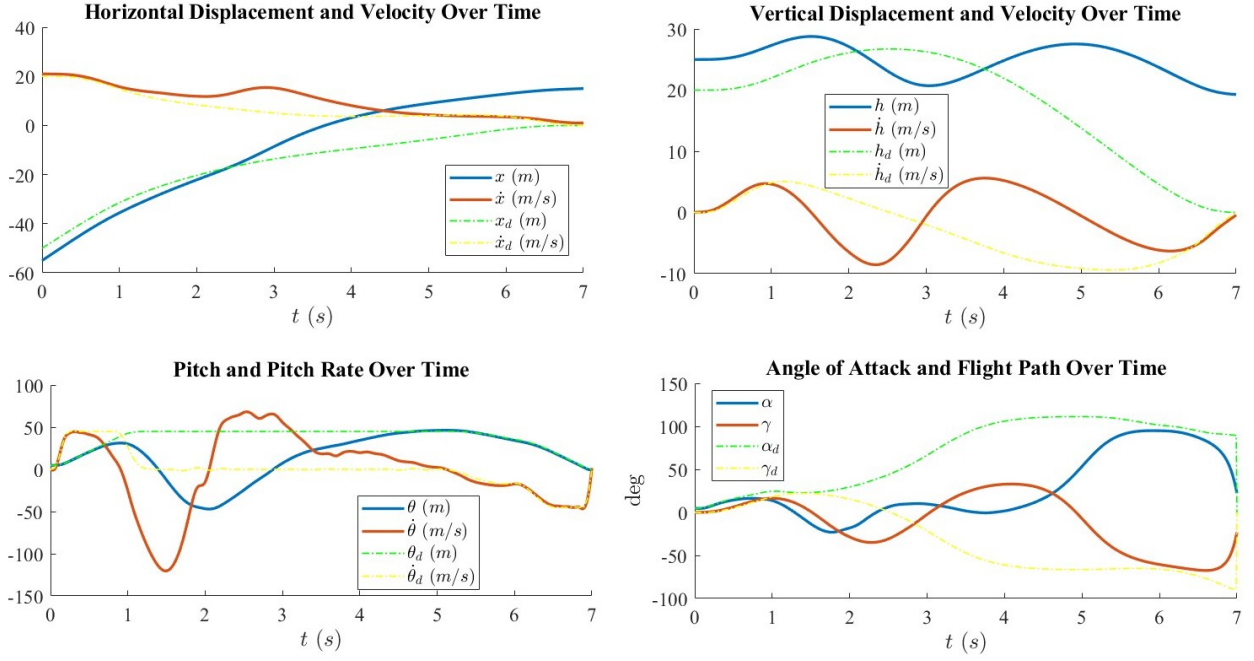


Figure C.19: Simulation 6: States vs desired states when initial position is disturbed.

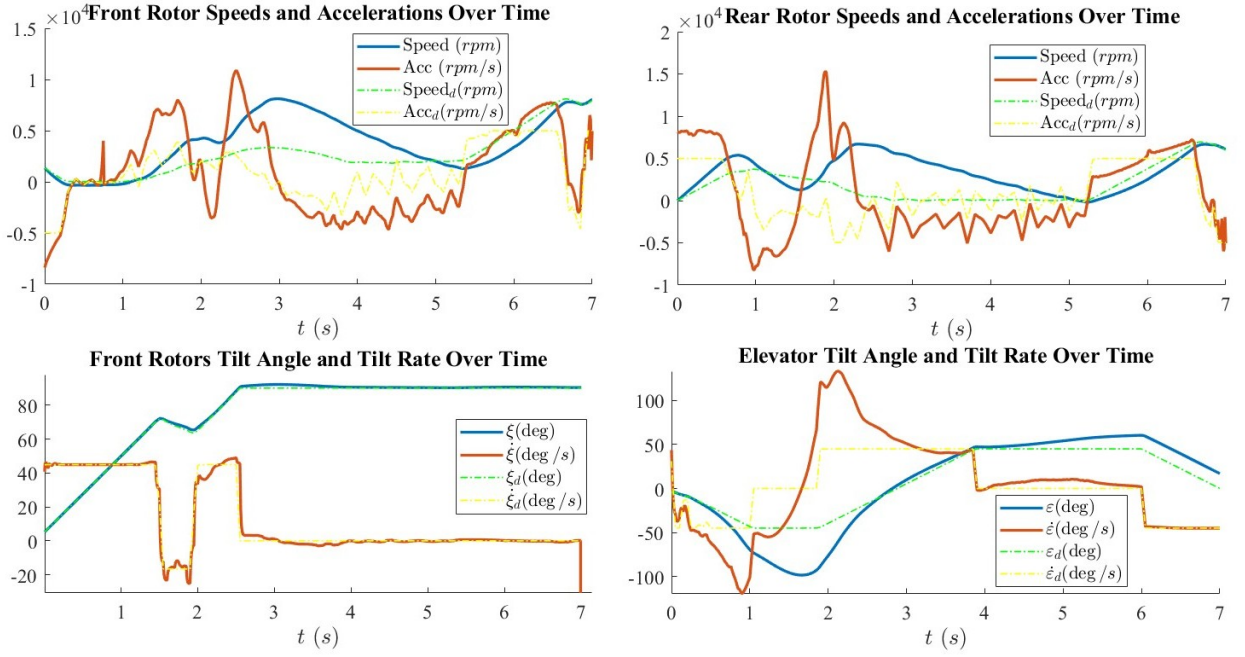


Figure C.20: Simulation 6: Inputs vs desired inputs when initial position is disturbed.

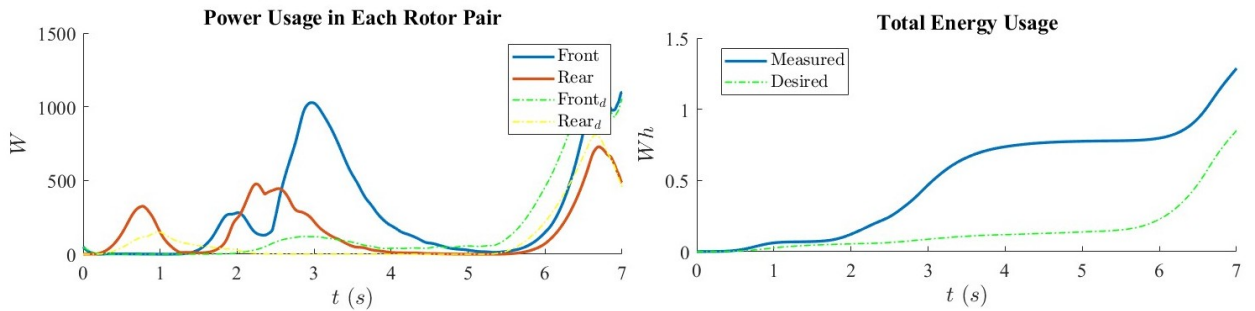


Figure C.21: Simulation 6: Comparison of additional energy consumption.

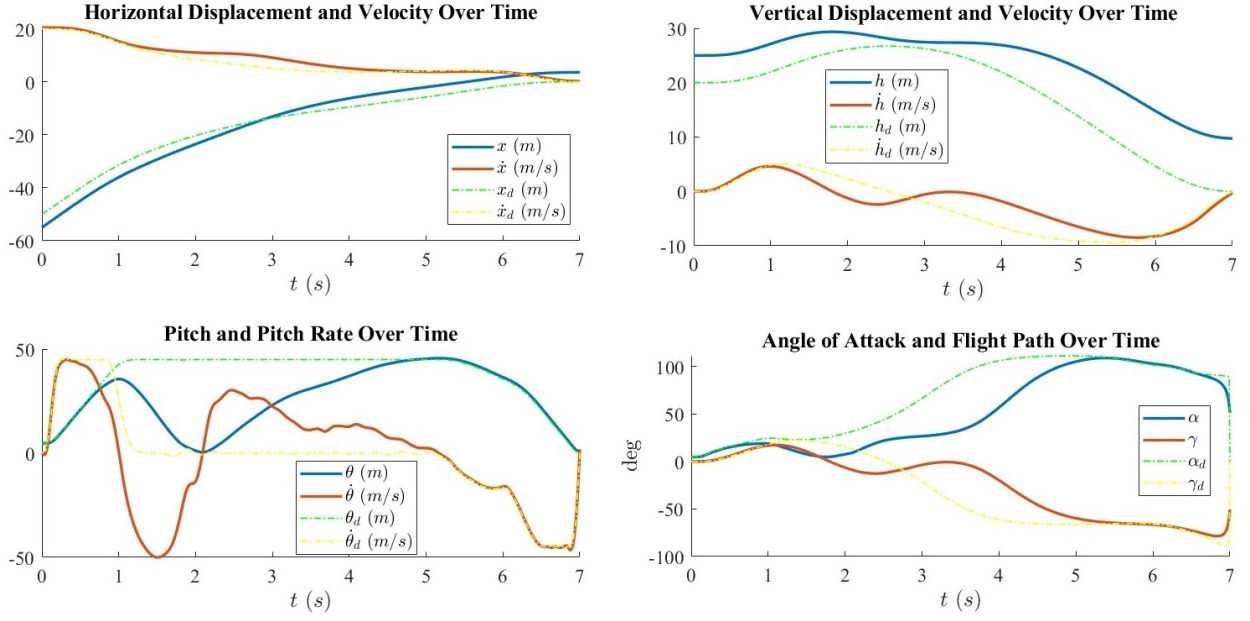


Figure C.22: Simulation 7: States vs desired states when initial position is disturbed.

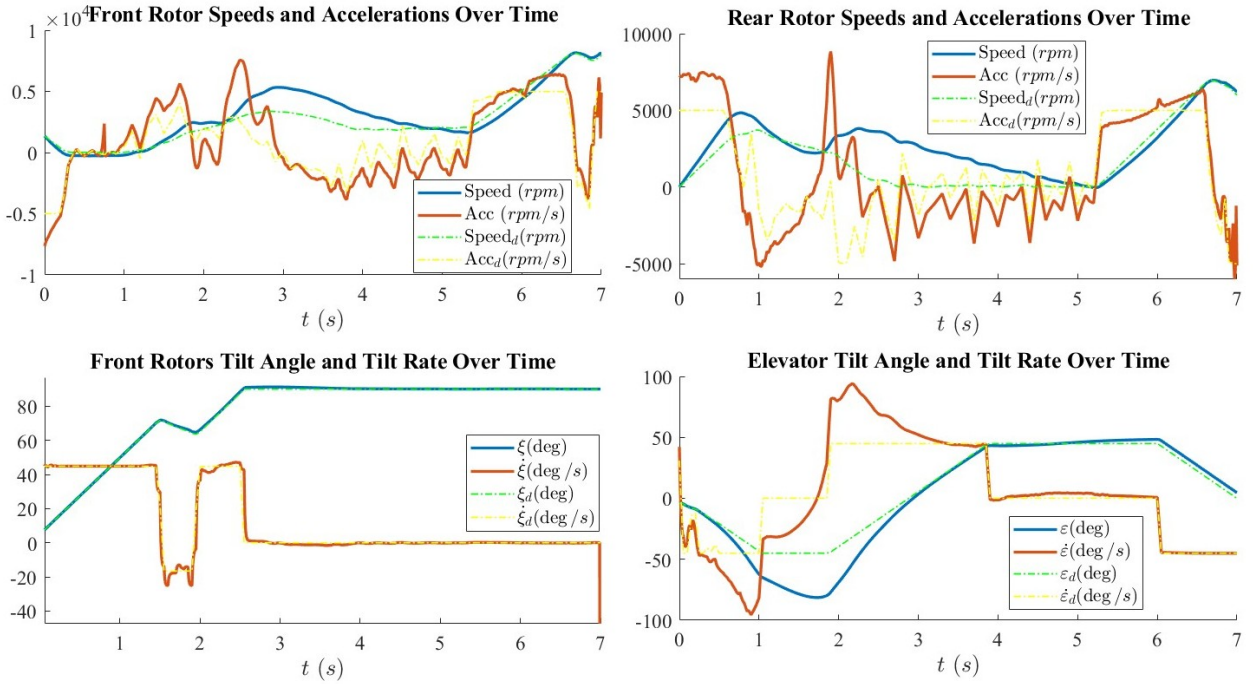


Figure C.23: Simulation 7: Inputs vs desired inputs when initial position is disturbed.

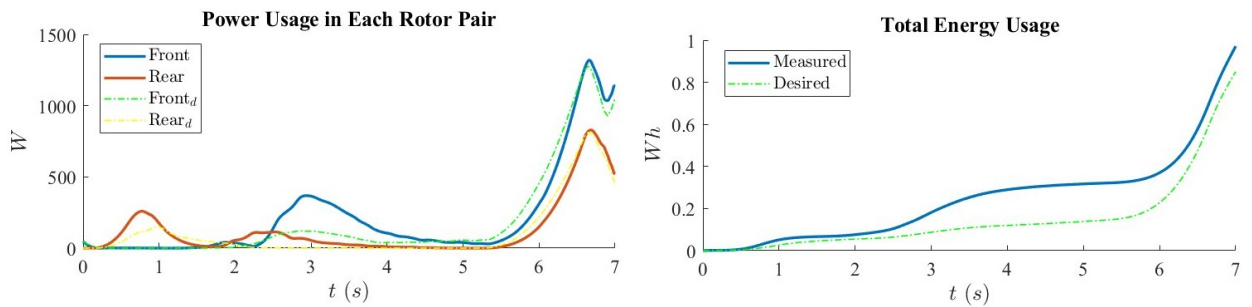


Figure C.24: Simulation 7: Comparison of additional energy consumption.

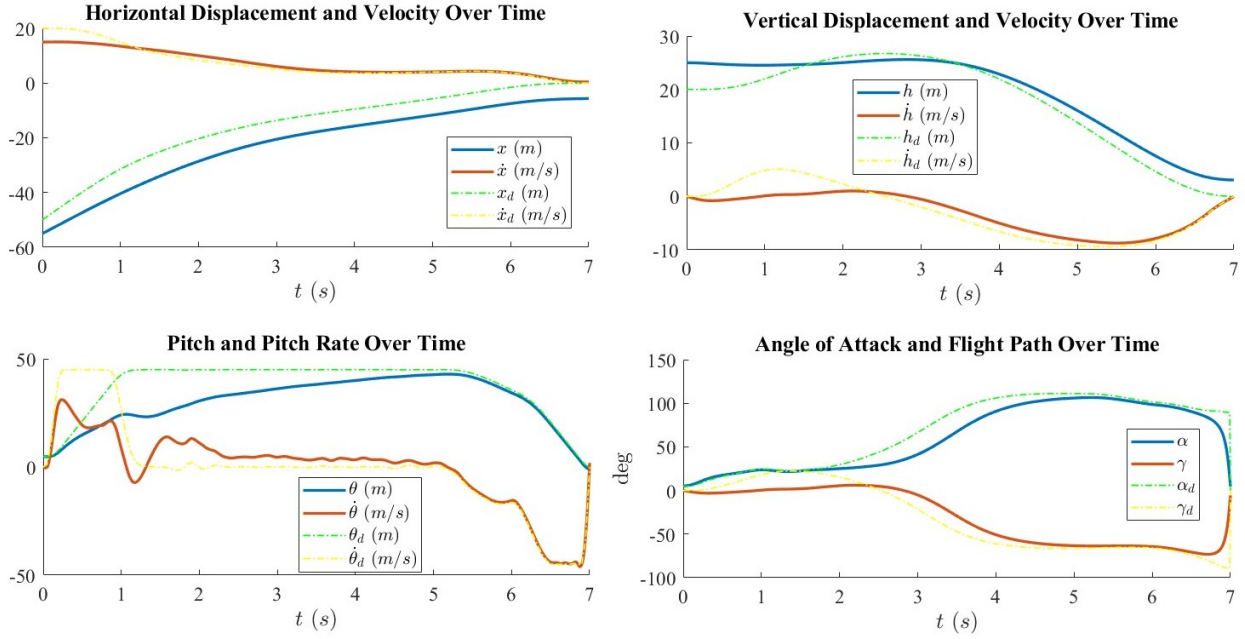


Figure C.25: Simulation 8: States vs desired states when initial position is disturbed.

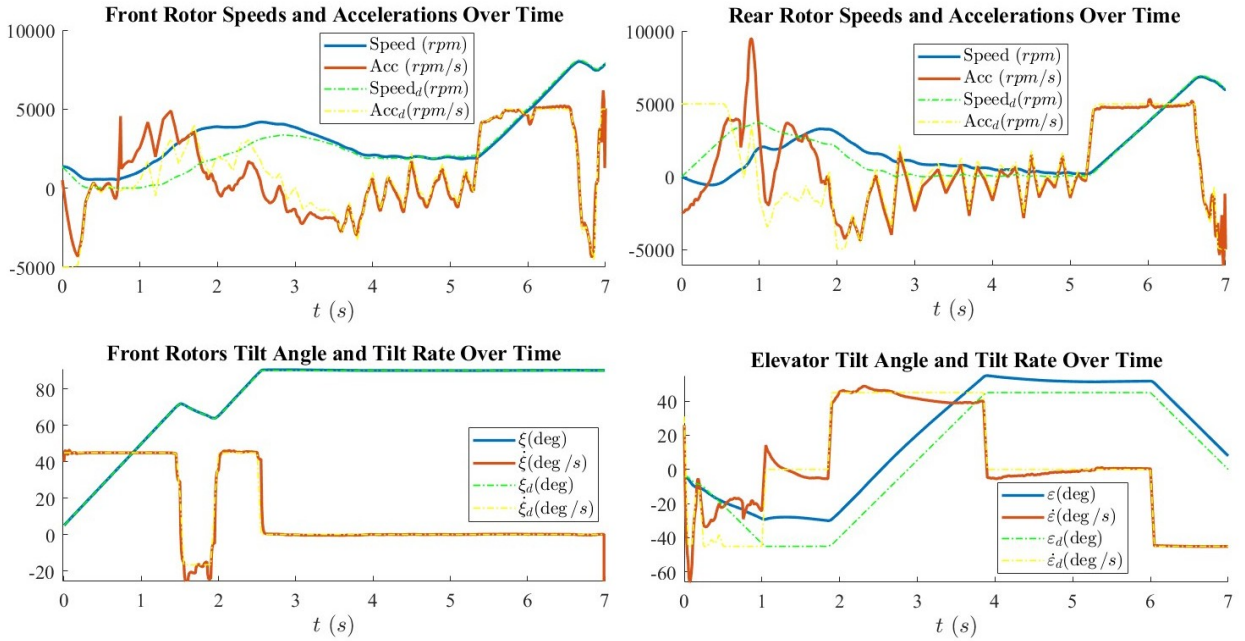


Figure C.26: Simulation 8: Inputs vs desired inputs when initial position is disturbed.

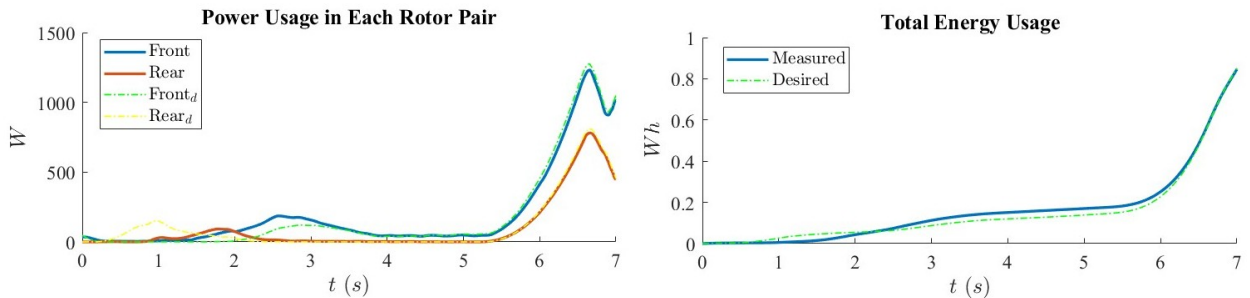


Figure C.27: Simulation 8: Comparison of additional energy consumption.